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Predicting bank defaults in Italy: A comparative analysis of conventional and machine learning approaches

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ABSTRACT

Although the probability of default (PD) modeling has reached a great maturity in both academia and business, for the Italian case we demonstrate that banks' available PD models would be misleading if today applied directly to Italian banks. We argue that what determines the PD of Italian banks, rather than the liquidity, are the return on assets (ROA), the financial leverage and the type of the bank. Furthermore, we demonstrate that the conventional approach dominates the more trendy machine learning (ML) and that model's performance could be used as a supervisory tool for retrospective analysis of the bank's position. This work stands out as the only study to consider state aid in defining bank default over the horizon period from 2010 to 2020, examining bank default in Italy across various types of banks, and employing both conventional and machine learning approaches, while also proposing an easy-to-handle three-variable model for predicting bank defaults.

1. Introduction

In this article, we address the Probability of Default (PD) of Italian banks due to the country's systemic importance, not only within Europe (Heimberger, 2020), but also globally as a member of the G7 and as a contributor to various initiatives on a global scale. We demonstrate that while there are widespread concerns in the markets regarding Italy's prospects and its debt service capacity, Daveri and Verona (2020) point out that the Italian government fulfilled its obligations despite exorbitant market rates and CDS spreads in 2009 during the sovereign debt crisis. The basis of Italy's resilience lies in its leading position in several sectors such as fashion (Mazzoni, 2020; Di Maria et al., 2023), pharmaceuticals (Ribbink, 2015; Ferriani et al., 2020), mechatronics (Dottori, 2021), food (Boccia et al., 2019; Fava et al., 2021), and tourism (Vergori and Arima, 2020; Bufalo and Orlando, 2023), among others. The Fortis-Corradini Index (FCI) indicates that "Italy ranked either first, second, or third worldwide in terms of foreign trade surplus for 932 products in 2012". Additionally, the FCI reveals that in 2012, only three countries (China, Germany, and the United States) surpassed Italy in the number of first, second, and third places in their trade balance globally (Fortis et al., 2016). S&P has confirmed the aforementioned solidity of the Italian banking sector despite the economic consequences of the pandemic. In fact, "on average, the probability of default for financial services companies is below the average for Italian companies in all sectors, standing at 1.4% and 2.6% over one and two years, respectively" (Standard&Poor's, 2020). In describing the Italian economy within the World Economic Forum initiative, Daveri and Verona (2020) underline the large share of small and medium-sized enterprises (SMEs) operating in Italy compared to other European countries. Consequently, many studies on the probability of default (PD) focus

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on Italian SMEs (e.g., Marotta et al. (2005), Fantazzini and Figini (2009), Provenzano and Arnone (2015), Ciampi (2015), Orlando and Pelosi (2020)). A recent comprehensive review by Ciampi et al. (2021) systematically analyzes the evolution of SME default prediction models over the past three decades, emphasizing the growing role of machine learning and alternative data sources in enhancing predictive accuracy. Less studied, conversely, is the segment of large companies in Ciampi and Gordini (2008), Altman et al. (2013), Micheli (2015).

To assess the Probability of Default (PD) of Italian banks, we identified a gap in the literature regarding the determinants of Italian bank default models. In fact, existing PD models may not be suitable for Italian banks due to their unique characteristics. To fill this gap, the study proposes an interpretable model with high predictive power, designed to be simple and transparent for rapid application. To achieve this, we developed two approaches: one classic and the other using a machine learning approach. We identified different models, among which the most performing one was labeled as "Pr05" and subsequently referred to as "the best model". This designation was based on several factors: it includes only statistically significant factors, has just four terms in the model, provides a pseudo R-squared value similar to alternative models, achieves one of the highest AUCs, and is not prone to multicollinearity. Specifically, this work contributes to the existing literature in several ways: (1) It stands out as the only study to consider state aid in defining bank default over the horizon period from 2010 to 2020. For instance, in 2017, the European Commission approved state aid totaling €5.4 billion for MPS and €4.78 billion for cash injections, along with €12 billion in state guarantees for the market exit of BPVI and Veneto Banca. This state aid, amounting to approximately €10.2 billion for that year alone, resulted in a 0.6 percent increase in public debt (International Monetary Fund (IMF), 2017), disproportionately affecting younger generations. (2) Additionally, this study is unique in its examination of bank default in Italy from 2010 to 2020 across various types of banks, including listed banks, unlisted banks, and small banks (such as LSIs (Bcc), Cooperative Banks (Popolari), Spa, Foreign branches, Listed, and (G-SII)), thereby reflecting the heterogeneity of the banking sector. (3) The research adopts both conventional and machine learning approaches, allowing for a comprehensive analysis of bank defaults. (4) Moreover, the study proposes a three-variable model, which is the most parsimonious in terms of explanatory variables, thus offering a more concise yet effective approach to predicting bank defaults. In fact, Calabrese and Giudici (2015) presents a model with 12 variables, while Ferriani et al. (2019) introduces one with 17 variables and another with 15 variables. The proposed parsimonious model is characterized by fewer data requirements and reduced computational complexity with practical advantages within both academic research and industry applications. These benefits contribute to enhanced model accessibility, operational efficiency, and interpretability.

Furthermore, Calabrese and Giudici (2015) does not include state aids in the distress definition due to their limited occurrence during 1996–2011, and Ferriani et al. (2019) does not consider state interventions because none of the Italian LSIs were affected by these events over the period considered (2008–2016). Conversely, our study incorporates state aid in defining bank default over the horizon period from 2010 to 2020.

Before introducing the model and delving into the heart of the analytical treatment, Section 2 offers readers a brief review of the literature on PD modeling in general, with specific applications for banks. This section also outlines the motivations behind this study. The dataset is described in Section 3, followed by the presentation of the proposed methodology and a comparison between advanced machine learning tools and conventional approaches in Section 4. The modeling results are summarized in Section 5, and Section 6 concludes by presenting policy implications for supervision and the estimated treatment effect of the BCC (Banche di Credito Cooperativo Casse Rurali ed Artigiane) reform in Italy.

2. Literature review and motivations

2.1. Probability of default modeling

Probability of Default (PD) modeling has its origins in the works of Beaver (1966), Altman (1968), Ohlson (1980). These initial efforts involved the application of discriminant analysis to relatively small datasets, often comprising only a few dozen observations. Modern studies, such as those in Moscatelli et al. (2020), have embraced machine learning (ML) techniques, processing datasets containing tens of millions of retail credit data points (for a comprehensive overview, interested readers can refer to Qu et al. (2019)). However, a challenge arises in this context. Machine learning techniques are rapidly gaining popularity, driven by regulatory requirements for transparent PD models from banks. Regulators are often hesitant to accept ML models for regulatory purposes, as discussed in European Banking Authority (EBA) (2021). A potential compromise has been proposed by the Bank of Spain, suggesting the use of conventional models for regulatory compliance and ML for validation, as discussed in Alonso and Carbo (2020).

PD models have been developed for various countries, including France (Jabeur and Fahmi, 2018), China (Chen et al., 2006; Liu et al., 2022), the UK (Altman et al., 2008), Japan (Tian and Yu, 2017), the EU (Bisogno et al., 2018), and Hungary (Kristóf and Virág, 2020). With reference to Italy, before launching the second Basel agreement (Basel II), the Basel Committee studied the existing results in PD modeling thus citing a study of the Italian Credit Register (CSFD) dating back to 1997 (BCBS, 2000, p. 110). We were unable to find the study as the reference does not contain sufficient details. Instead, we found an equivalent document by Laviola and Trapanese (1997). However, we suspect that these are two different documents as they present a different number of observations: 1885 in the Italian CSFD study from BCBS (2000) and 1274 in Laviola and Trapanese (1997).

2.2. PD models and macroeconomic indicators

Traditional probability of default (PD) models have heavily relied on macroeconomic variables — such as interest rates, stock returns, and GDP growth — to forecast corporate defaults (Das et al., 2007; Koopman et al., 2009; Duffie et al., 2009; Lando and Nielsen, 2010; Azizpour et al., 2018). Mileris (2012) develops a statistical model linking macroeconomic indicators — such as GDP, inflation, and money supply — to changes in banks' loan portfolio credit risk across 22 EU countries, achieving high

predictive accuracy for future non-performing loan levels. Agrawal and Maheshwari (2014) show that, for Indian listed firms, stock market sensitivity is positively associated with default probability, while sensitivity to inflation (CPI) is negatively associated, based on logistic regression and discriminant analysis. Bhimani et al. (2013) demonstrate that incorporating non-financial information alongside financial and macroeconomic variables significantly enhances the prediction of bank loan default timing, particularly for small and medium enterprises. Simons and Rolwes (2018) apply a macroeconomic-based model to Dutch firms and find that while GDP growth and oil prices significantly influence default rates, a stress scenario involving two quarters of zero GDP growth has a limited impact on predicted defaults.

2.3. PD models specific to the banking sector

PD models tailored for banks are relatively scarce. Pertinent reviews can be found in Kumar and Ravi (2007) and Citterio (2020). This scarcity is due to banks often falling under the category of a so-called "low default portfolio (LDP)", as discussed in Penikas (2020). In these cases, the default rate — indicating the proportion of the defaulting class within the entire set of observations — is nearly negligible.

However, the Lehman Brothers case serves as a reminder that even a single default can have significant repercussions. The collapse of Lehman Brothers could have triggered a bank run on the entire US banking system and led to its complete collapse within just two days, as noted by Krugman (2009). This event prompted the state deposit insurance limit to be more than doubled, rising from USD 100k to USD 250k within a mere two days after the Lehman incident. This example underscores how a seemingly insignificant fraction of bank insolvencies can hold immense importance for national and sometimes even global financial stability. This underscores the importance of continuing to develop specialized PD models for a specific segment of borrowers—banks.

Among the various studies focusing on bank defaults and distress, several notable examples include Bräuning et al. (2020) and Durand et al. (2021) for the EU, (Yuksel et al., 2015) for Turkey, (Shrivastava et al., 2020) for India, (Kočenda and Iwasaki, 2022) for Japan, (Cole et al., 2020; Manthoulis et al., 2021) for the USA, (Obeid, 2021) for the Persian Gulf region, (Saliba et al., 2023) for BRICS (Brazil, Russia, India, China, and South Africa) countries, Cheong and Ramasamy (2019), Orlando et al. (2022), and Kristóf (2021) for others. Recent studies by Malikkidou and Strohbach (2025) and Hu et al. (2025) highlight the effectiveness of machine learning, particularly Random Forests, in predicting bank distress. Malikkidou and Strohbach emphasize the role of sampling strategies, feature selection, and profitability and solvency indicators in improving early warning systems, while Hu et al. show that incorporating time dynamics enhances predictive accuracy, though model performance weakens under severe stress conditions. Additionally, alternative sources of borrower-level information, such as transaction and account activity data, have been shown to significantly enhance early warning signals for default risk, particularly among SMEs and individuals (Norden and Weber, 2010).

Significant contributions to banking PD modeling are associated with the RiskCalc model developed by Moody's. The document by Kocagil et al. (2002) covers banks in the USA, while an equivalent model for banks globally became available only a decade later, as presented in Moody's Analytics (2016).

Interestingly, despite the Russian banking system not being the most developed globally, Russian bank data have garnered attention as a valuable source for PD model development. Notable examples include scholars from Finland, such as Fungáčová and Solanko (2009), Fungáčová et al. (2010), Fungáčová and Weill (2013), Fungáčová et al. (2020); The Netherlands, as seen in Karas and Vernikov (2019); Switzerland, with contributions from Goncharenko et al. (2022); and even Russia itself, as explored by Shibitov and Mamedli (2019). This growing interest is primarily due to the Central Bank of the Russian Federation publishing detailed monthly financial statements of local credit institutions. Similarly, such financial disclosures for American banks also exist, although they seem to pose more challenges when it comes to processing:

- For existing banks (Council, 2022), though earlier it was available at Federal Reserve Bank of Chicago (2022);
- For failed ones (F.D.I. Corporation, 2022).

For a synthetic overview of the main contributions to the literature on bank default estimation, see Table A.7 in Appendix A.

2.4. PD models specific to the Italian banking sector

Surveys focusing on Italian banks within Italy remain scarce with few exception. Before the banking reform, Calabrese and Giudici (2015) demonstrated that, using a sample of 783 Italian banks (1996–2011), only firm-specific factors and regulatory capital were significant in explaining bank failures. The working paper authored by Ferriani et al. (2019), while it does have certain limitations, remains the most comprehensive study to date on the matter. To start, the paper employs non-public (supervisory) data to assign a default (financial distress) flag. Furthermore, the study's scope is confined to less significant banks, thus omitting the largest ones. The research's innovation lies in the introduction of a co-operative banks (BCC) dummy variable by Ferriani et al. (2019). Italy executed a distinctive reform six years prior, on February 16, 2016, which entailed the consolidation of all co-operative banks under a single supervisory entity, as outlined in EACB (2016). These 367 Italian mutual banks together constitute an SSM (Single Supervisory Mechanism) significant banking group, standing as the third-largest Italian banking institution and the foremost in terms of capital, as reported by EACB (2016).

However, the reform turned out to be not only complex but also remarkably distinctive, not just within Italy but globally as well. Roughly a year after the reform's initiation, it became apparent that a singular entity could not effectively manage the

operations of all member cooperative banks. This realization led to the emergence of two distinct banking groups: the Gruppo Bancario Cooperativo Iccrea and Cassa Centrale Banca.

Of these, the former holds the larger number of member entities, totaling 147 as of September 2, 2022. This ensemble includes 120 cooperative (BCC) banks, 21 controlled entities (società controllate), and 6 mutual organizations (società collegate) under the parent Iccrea bank, according to details from Iccrea (2022).

Conversely, the latter is of smaller size and specifically focuses on rural banks. As of July 1, 2022, this group comprises 69 banks, as detailed on page 16 of CCB (2022). Additionally, apart from the two groups headed by the parent S.P.A. banks (Iccrea and Cassa Centrale), an independent group emerged at the end of 2020 in the northern region of Italy. This group is referred to as the Institutional Protection Scheme (IPS) of the South Tyrolean Raiffeisenkassen, as discussed in IRU (2020). With 39 members, this group operates without a single superior bank, unlike the aforementioned two groups. For an overview of the main contributions to the literature on bank default estimation specific to Italy, see Table A.8 in Appendix A.

2.5. PD models specific to small banks and cooperative groups

It is important to note that the term 'entity' is not accurately applicable when describing cooperative groups, as the reform did not intend to establish new legal entities or banking holdings, as explained by FitchRatings (2022b). Instead, the cooperative groups are supervised by banks that share the same name. However, smaller member banks participate in a role equivalent to shareholders within these groups. Consequently, these minor members can exert partial control over the group to balance decision-making alongside the major (anchor) bank of the group. This unique dual management and control structure within the group distinguishes it from conventional financial practices worldwide. To some extent, it bears resemblance to the cyclical ownership structures often found in industrial conglomerates in Japan.

The question that arises at this juncture is whether a new PD model needs to be developed or if existing solutions, such as the well-established (Moody's Analytics, 2016) model crafted from a dataset encompassing 90 countries, can be utilized. Anticipating the obtained results, we demonstrate that Italy possesses distinct characteristics that are not adequately addressed by either the aforementioned Moody's model or the closest benchmarks outlined in Ferriani et al. (2019) and Penikas (2022).

3. Data analysis and preparation

This section comprises five distinct parts. The initial segment elucidates how the definition of defaults is applied and their exploration, followed by the structure of the data. Subsequently, the classification of outliers is expounded upon, and the final part is devoted to a preliminary data analysis.

3.1. Definition of default

Within this section, we, give a definition of default used for the assignment of the default flag. Before describing the assignment of the default flag it is necessary to provide the definition adopted for the purposes of this work to classify banks in "default" or "in bonis" status (or performing bank). Although from 1 January 2021, the new definition of default came into force, this new definition concerns how banks must classify customers for prudential purposes but this clarification does not apply to the classification process of credit institutions (banks) in default.

Considering that bank failures are not so frequent events and assuming that as reported by Durand et al. (2021) "there is no official list of failed banks in Europe" and that there is no official definition of default to be adopted for the classification of banks in default - except for the extraordinary administration and provisional management procedures published by the Bank of Italy - and also considering that the legislator identifies banking crises by degrees of severity and intervention. Therefore for the purposes of our work, we have chosen to assign the status of banks in "default" to those institutions, which in the temporary horizon between 2010 and 2020, have been subject to at least one of the following procedures: i. the merger or acquisition by a third party; ii. bail-out and/or "state aid"; iii. bail-in, requiring banks to recapitalize and absorb losses from shareholders, depositors and creditors; iv. intervention by the Interbank Deposit Protection Fund (FITD); v. extraordinary administration (intervened by the ECB or Bank of Italy); vi. compulsory administrative liquidation (or winding up).

Article 107 of the Treaty on the Functioning of the European Union defines "state aid" as *'any aid granted by a Member State or through State resources in any form whatsoever which distorts or threatens to distort competition by favoring certain undertakings or the production of certain goods [. . .], in so far as it affects trade between Member States'* (European Commission, 2016). This Notice clarifies the different constituent elements of the notion of State aid: the existence of an undertaking, the imputability of the measure to the State, its financing through State resources, the granting of an advantage, the selectivity of the measure, and its effect on competition and trade between Member States.

To assess the scale of state aid intervention between 2008 and 2017, the Commission approved aid to the financial sector amounting to €1459 billion (capital-like aid) and €3659 billion (liquidity aid). These measures constituted taxpayer-funded bank bailouts (European Court of Auditors, 2019).

Due to the ambiguity in the definition of default, researchers often prefer early-warning-system models, which allow for an expanded definition (or a "wider definition of possible distress events", as in Ferriani et al. (2019)) of a "banking crisis". This broader interpretation helps to increase the number of observations available for rebalancing the studied classes.

Table 1
Default records.

Name of the Bank classified as "Default"	Year of Default	Reason of Default
1.Bpm Banca Popolare di Milano S.C.A.R.L	2016	Merger
2. Carige Cassa di Risparmio di Genova e Imperia	2019	Ex.Ad.
3. Mps Banca Monte dei Paschi di Siena	2017	Bail-Out
4. Ubi Unione di Banche Italiane S.p.A	2020	Merger by Incorporation
5. Banco Popolare Soc. Coop.	2016	Merger
6. Banca di Sardegna	2020	Merger by Absorption
7. Banca Popolare di Spoleto	2013	Ex.Ad.
8. Veneto Banca	2017	CAL
9. Bene Banca Bene Banca Credito Coop. di Bene Vagienna	2013	Ex.Ad.
10. Carife Cassa Di Risparmio di Ferrara S.p.A	2012	Ex.Ad.
11. Bcc di Alberobello e Sammichele di Bari	2013	Ex.Ad.
12. Bcc di San Francesco Banca San Francesco Credito Cooperativo Canicattì	2012	Ex.Ad.
13. Banca Popolare di Bari S.c.p.a.	2019	Ex.Ad.
14. Bcc di Cittanova Banca di Credito Cooperativo di Cittanova S.C.	2017	Ex.Ad.
15. Banca di Credito Peloritano	2020	Ex.Ad.
16. Invest Banca S.p.A.	2020	Ex.Ad.
17. Banca del Sud S.p.A.	2019	Ex.Ad.
18. Istituto per il credito sportivo	2012	Ex.Ad.
19. Banca delle Marche S.p.A.	2012	Ex.Ad.
20. Banca Popolare di Vicenza S.p.A.	2016	CAL
21. BCC del Veneziano	2011	Ex.Ad.

Note: *"Year of Default" is the year where we assigned the default flag, while "Reason of Default" corresponds to the reason of default. "Ex.Ad." stands for extraordinary administration and "CAL" for compulsory administrative liquidation.*
The assignment of the default flag was reported in the year of the communication by the authorities of one of the events reported under our definition of default, (e.g. extraordinary administration 2019, financial statement available 2019, then default flag 2019). In those cases where the financial statements were not drawn up in the year in which the communication by the authorities took place, then the flag default was assigned to the last available financial statements before the communication (e.g. 2019 extraordinary administration, financial statement not available 2019, then default flag 2018).

The default records in Table 1, based on the definition of default provided earlier, were compiled by consulting the *Relevant provisions relating to entities subject to supervision* section on the Bank of Italy website, along with related historical documents and communications from the Ministry of Economy and Finance (MEF) published in the *Gazzetta Ufficiale della Repubblica Italiana*.

The assignment of the default flag corresponds to the year in which one of the events falling under our definition of default was officially communicated by the authorities (e.g., extraordinary administration in 2019, financial statements available for 2019, hence default flag set to 2019). In cases where the financial statements were not prepared for the year of the communication, the default flag was assigned to the most recent available financial statements preceding the communication (e.g., extraordinary administration in 2019, no 2019 financial statements available, hence default flag set to 2018). This approach avoids assigning a default flag to financial records with zero or missing values.

3.2. Description of data

The banking sector in Italy is notably diverse, comprising a range of companies differing in terms of size, geographic spread, operations, core business, and ownership structure. Given the infrequency of default cases, the preference was to select a sample with substantial diversification. This strategy aimed to accomplish two objectives: first, to mirror the Italian banking landscape, and second, to yield an extensive dataset. Regarding sample construction, criteria used to select or dismiss a bank, and sample constriction by data availability see Appendix B

To provide appropriate and accurate context for the results derived from this study, certain clarifications are necessary. These clarifications serve to contextualize the outcomes effectively, considering the potential limitations if we solely adhere to the reference sample.

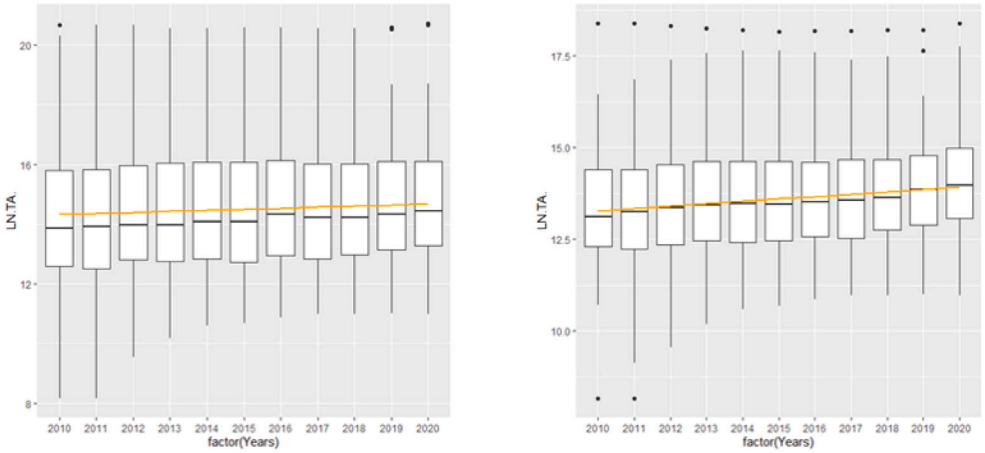


Fig. 1. The logarithm of total assets (LN_TA) is depicted for each year across the entire dataset, which includes 27 consolidated financial statements and 73 individual financial statements (shown on the left). The distribution for individual financial statements alone is presented on the right.

Due to various reforms, such as the implementation of the International Financial Reporting Standards (IFRS) system, which replaced the International Accounting Standards (IAS) in 2001, budget statements have changed over time, resulting in variations in some budget indicators. The variables under examination are easily identifiable in financial statements because they are macro-items historically utilized in such documents, and they remain consistent regardless of the type of bank being considered. For example, these variables include ROA (PROFIT/TOTAL ASSETS), PN/TA (NET EQUITY/TOTAL ASSETS), and so on.

The dataset comprises 1059 bank financial statements (income statements and balance sheets), assembled through a rigorous manual data entry process, where each numerical value was individually entered and cross-checked against the original PDF documents to guarantee the highest standards of data accuracy. Subsequently, these financial statements were reconciled and reclassified to construct 72 performance indicators, which will serve as explanatory variables within the predictive models. A total of one hundred banks were analyzed in this study, encompassing both group leaders of banking groups and individual banks. The observation period spans from 2010 to 2020. Among these, there are 27 banking groups or institutions characterized by widespread shareholding or controlled by other companies or entities. Additionally, there are 10 cooperative banks and 63 credit institutions affiliated with cooperative credit banks and rural banks within the sample.

Within the 100 observed banks, the financial statements of 27 banks were consolidated, while the financial statements of the remaining 73 were at the individual (solo-level) level. In terms of representativeness of the sample, among the above mentioned 100, the 5 biggest banks Intesa Sanpaolo, UniCredit, Banco Bpm, Banca MPS, BPER Banca and BNL constitute approximately 54.45% of the market share of the entire Italian banking system as of 2021 (Mileiko, 2023). Moreover, those banks of have been identified by the Bank of Italy as systemically important institutions (Bank of Italy, 2023; European Banking Authority, 2023). Regarding the BCCs, they represent 52.31% of the banks operating in Italy and 19.21% of Italian bank branches (Federcasse, 2021).

Regarding the distribution of total assets, 25% do not exceed €384.95 million, 50% do not surpass €1.46 billion, and in 25% of cases, total assets exceed €8.89 billion. Indeed, the distribution comprises a substantial number of very small banks along with a limited number of large banks.

To offer a comprehensive perspective on the variation in company size within the sample, two graphs are presented in Fig. 1. The left chart in Fig. 1 displays the logarithm of total assets for each year across all financial statements within our dataset, encompassing 27 consolidated "BC" financial statements and 73 individual "BI" financial statements. The distribution for individual financial statements alone is depicted on the right side of Fig. 1. Consequently, it is noted that 75% of the values from individual financial statements remain below €2.2 billion (approximately 15 units in terms of the natural logarithm value of total assets).

This characteristic prompts us to initiate our analysis by first considering the entire dataset (comprising both BC and BI financial statements) encompassing 100 banks. Subsequently, we narrow our focus solely to the subset of 73 individual financial statements (BI). This approach is undertaken to assess whether variations in asset size or the prevalence of certain balance sheet items might yield disparate outcomes within the models, affecting both performance and the identification of significant variables.

For our subsequent analysis, we divide the data into two distinct subsets:

- DS(B100) Comprising both types of financial statements for a total of 100 banks, including consolidated and non-consolidated statements. This subset encompasses 21 instances of default.
- DS(B73) Derived from non-consolidated reports of 73 banks, incorporating 11 occurrences of default.

It is important to note that the larger DS(B100) sample is equally significant for the development of the PD model. To ensure accuracy, double counting due to the utilization of both consolidated and non-consolidated financial statements has been appropriately circumvented. Consolidated reports are exclusively considered for banks where unconsolidated financial statements are unavailable.

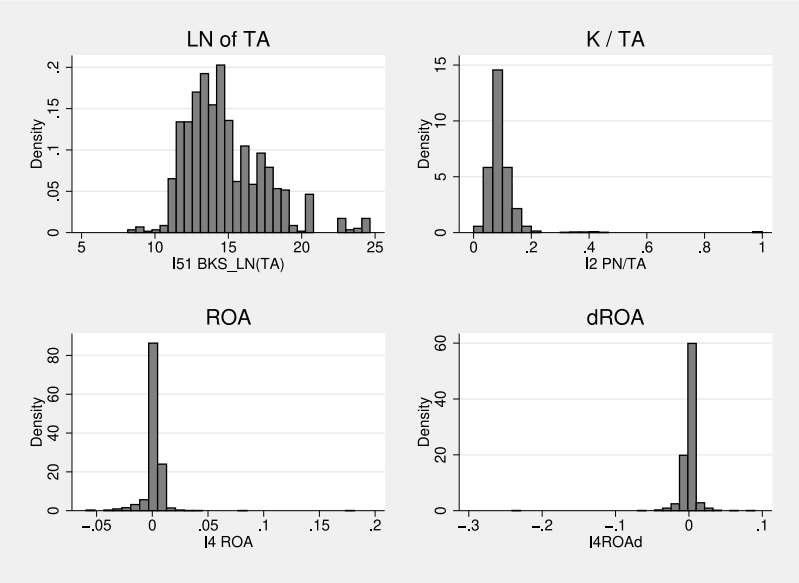


Fig. 2. There is need to filter for outliers.

3.3. Prediction of default

To predict the state of *one year before default* for Italian banks starting from the default flag assignment (which is the cornerstone of any PD model development), we first take the standard approach and then compare the current year's default with the previous year's financial data. The financial data for the current year undoubtedly reflects the deterioration in creditworthiness in the event of default this year. This is because financial results at a given moment are useful to test the goodness of the forecast. We begin by adopting the standard approach, which involves comparing the current year's default status with the financial data from the preceding year. In this context, the financial data for the present year inherently reflects the creditworthiness deterioration should a default occur in the current year. This is due to the fact that financial outcomes at a given point in time serve as a valuable measure for assessing the accuracy of our forecasts.

3.4. Classification of outliers

The initial step in our process is data cleansing. Given that we employ logarithms of the variations, our first task is to eliminate observations with zero values.

Subsequently, our focus shifts to identifying the most suitable candidates for the PD drivers. Upon reviewing Fig. 2, it becomes evident that outliers are prevalent, particularly in relation to the profitability ratio, ROA (Return on Assets), and the change in ROA. Notably, these values are influenced by the distinctive nature of Italian banks.

Having established the aforementioned details, we proceed to define outliers based on the criteria defined in Table 2.

Given that a few banks possess more than one default flag, the overall count of outliers is 14 (in contrast to the 21 indicated by Table 2). Consequently, Fig. 3 illustrates the distribution of data post the removal of outliers.

In total, we identify 2.2% of the 946 bank-year observations as outliers. This proportion is relatively small and falls below the conventional confidence threshold of 5%. This emphasizes the significance of the presented outlier criteria list. It is intentionally designed to prevent excessive attrition within our sample of defaulted observations. Striking this balance is crucial as further reduction could impede our ability to effectively train (fit) a default prediction model (Beamish and Hasse, 2022).

3.5. Preliminary analysis of data

The observed default rate within the considered sample is 2.6% annually, aligning with what is referred to as a "low default portfolio" (LDP). It is noteworthy that Fig. 4 (right) suggests the presence of a bimodal distribution for the default rate. This observation indicates a notable degree of default correlation within the LDP segment. Upon examining the trajectory of Italian GDP growth, an interesting hypothesis emerges: a lag of four years could potentially hold relevance for the PD of Italian banks. However, as evidenced in Fig. 5, this trend is not distinctly pronounced. Subsequently, incorporating such a GDP growth lag as a PD driver does not yield a statistically significant coefficient. It is important to acknowledge the interplay between bank insolvencies and the prevailing economic circumstances, mirrored in the GDP growth figures. Economic downturns wield an impact on both bank balance sheets and the GDP. A cyclical mechanism comes into play: economic crises stimulate bank insolvencies, which in turn curtail the

Table 2
Classification of outliers.

Criteria	# of All	# of Def.
1. ROA (I4ROA) too low (below -6% p.a.) or too high (above +6% p.a.)	2	0
2. ROE (I1ROE) too low (below -100% p.a.)	2	0
3. Operating profit to brokerage margin (I43RGMID) too low (below −2) or too high (above +2)	5	0
4. Exorbitant total capital ratio (I11TCR) above 90% to capture the risk of assets	5	2
5. Exorbitant leverage ratio (I2PNTA) above 90% to capture (non adjusted) risk	3	0
6. Extreme change in ROA (more than 5 pp. p.a. in absolute terms)	4	0
Total	21	2

Note. *"# of All" is the number of outliers with respect to the entire sample (there are 2.2% of total 946 observations), while "# of Def." corresponds to default data points only (there are 8.0% of total 25 cases).*

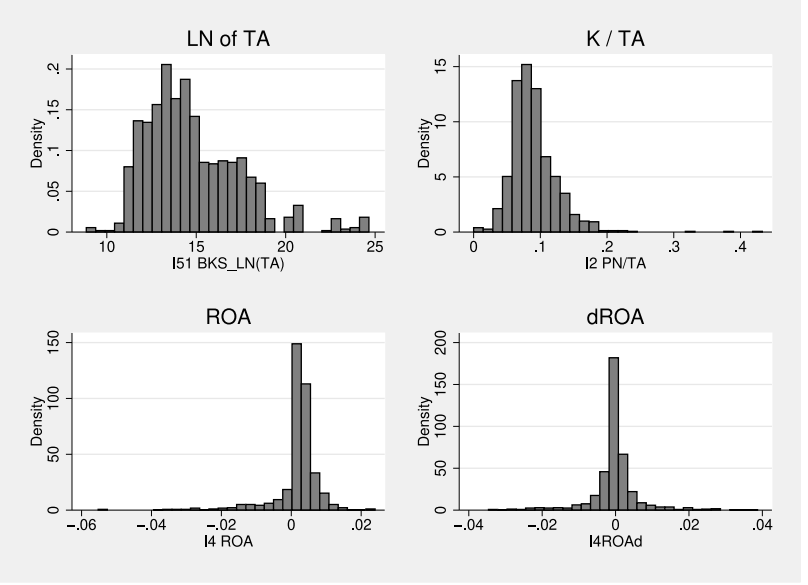


Fig. 3. After filtering for outliers the distributions approaches bell-shaped (though not Gaussian) form.

availability of credit financing necessary for swift recovery. Given the intricate nature of the relationship between macroeconomic variables and bank failures, and the need for a customized approach, especially considering the interwoven presence of the economic cycle within financial data (e.g., Saliba et al. (2023) demonstrates that default is country-specific), we have deliberately chosen to exclude these variables from our model. We have opted to defer the examination of this complex matter to future research. Furthermore, as indicated by Calabrese and Giudici (2015), the inclusion of macroeconomic factors such as GDP growth rate, inflation rate, unemployment rate, and interest rate has revealed that none of these variables are statistically significant.

A brief examination of the conventional PD drivers in Fig. 6 implies that an Italian bank exhibits greater stability when (a) it maintains a relatively modest size (addressing the 'too big to fail' issue); (b) its capital holdings are substantial; (c) it achieves higher levels of profitability; and (d) its profits experience notable growth. These trends correspond to well-established observations regarding banking sustainability. We have also explored the possibility of quadratic patterns, but it is evident from Fig. 6 that the parabolic trends largely coincide with the linear ones. This consideration led us to omit squared factor values from our PD model.

4. Methodology

This section outlines the overall methodological framework adopted in the study, providing a structured overview of the steps taken to develop, validate, and interpret probability of default (PD) models for Italian banks. As summarized in Fig. 7, the research design follows a sequential process that begins with the construction of a comprehensive dataset combining supervisory information,

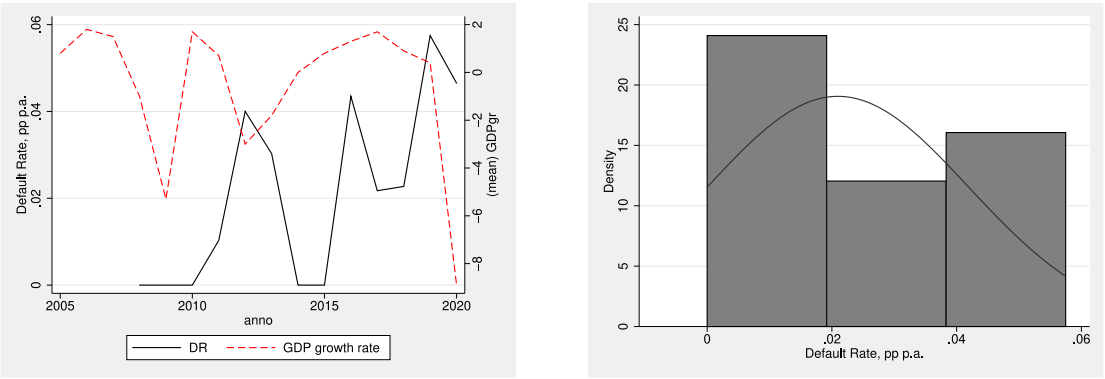


Fig. 4. The world financial crisis of 2007–09 triggered defaults among Italian banks (left). The bimodal DR distribution signals for the presence of positive default correlation (right).
Note: DR - the default rate (pp. per annum, p.a.).

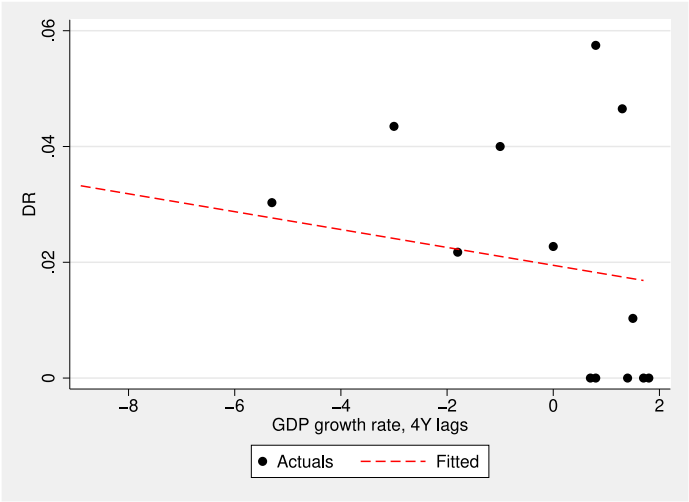


Fig. 5. Higher pace of economic growth is associated with lower bank defaults in Italy, though the path is not quite robust.
Note: DR - the default rate (pp. per annum, p.a.).

financial statements, and macroeconomic variables. This is followed by a preprocessing stage, where data are cleaned, winsorized, and adjusted for class imbalance prior to the division into training, validation, and test sets.

Subsequently, we implement and compare two broad families of models: (i) conventional econometric approaches, including logit, probit, linear probability, and panel specifications; and (ii) machine learning techniques, such as CART, random forest, and penalized regressions (LASSO, Ridge). For each family, we evaluate model estimation, calibration, and diagnostic properties. The models are then subjected to a comprehensive validation exercise based on K-fold cross-validation, AUC-ROC, Cohen's kappa, Youden's J, Brier score, F1, and temporal/sectoral stability. We complement this with stress scenario analysis, where adverse macro-financial conditions are simulated to test the resilience of predictions.

The subsequent stage involves interpretation and robustness checks, where we contrast marginal effects and coefficients from conventional models with feature importance and SHAP values from machine learning. Additional robustness exercises — such as placebo tests, alternative specifications, and sub-sample analyses — are used to confirm the stability of results. Finally, the findings are synthesized and linked to regulatory and policy implications, with particular emphasis on the early-warning potential of the models in the unique context of Italy's banking sector.

4.1. PD model development and validation

Having thoroughly analyzed and prepared the data, the subsequent step involves modeling through the utilization of both widespread machine learning (ML) techniques and conventional tools. In the domain of ML, we have harnessed (a) the least absolute shrinkage and selection operator (LASSO) and ridge regression to identify the most influential PD drivers, (b) classification and regression tree (CART), (c) random forest, and (d) GLM regression.

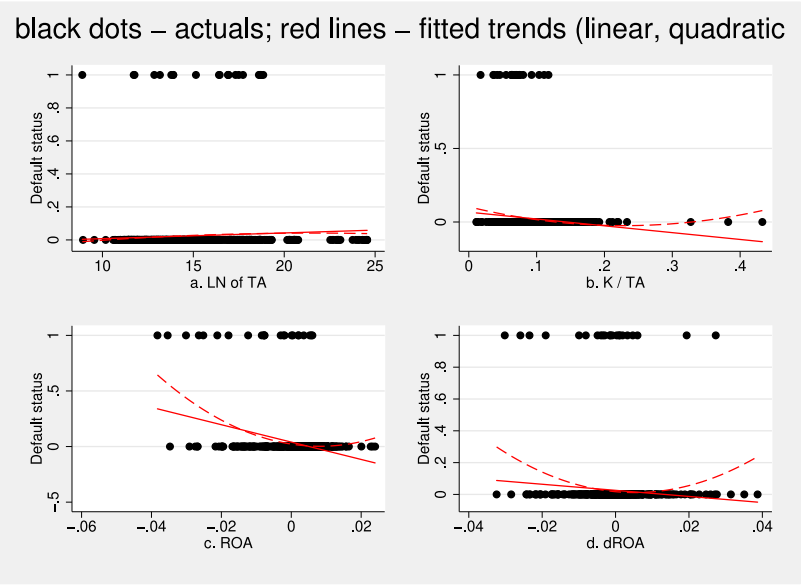


Fig. 6. Higher capital and higher profitability is associated with low PD of Italian banks.
Note: black dots stand for actual observations; solid red line is the linear fit, while the red dashed line is the parabolic trend.

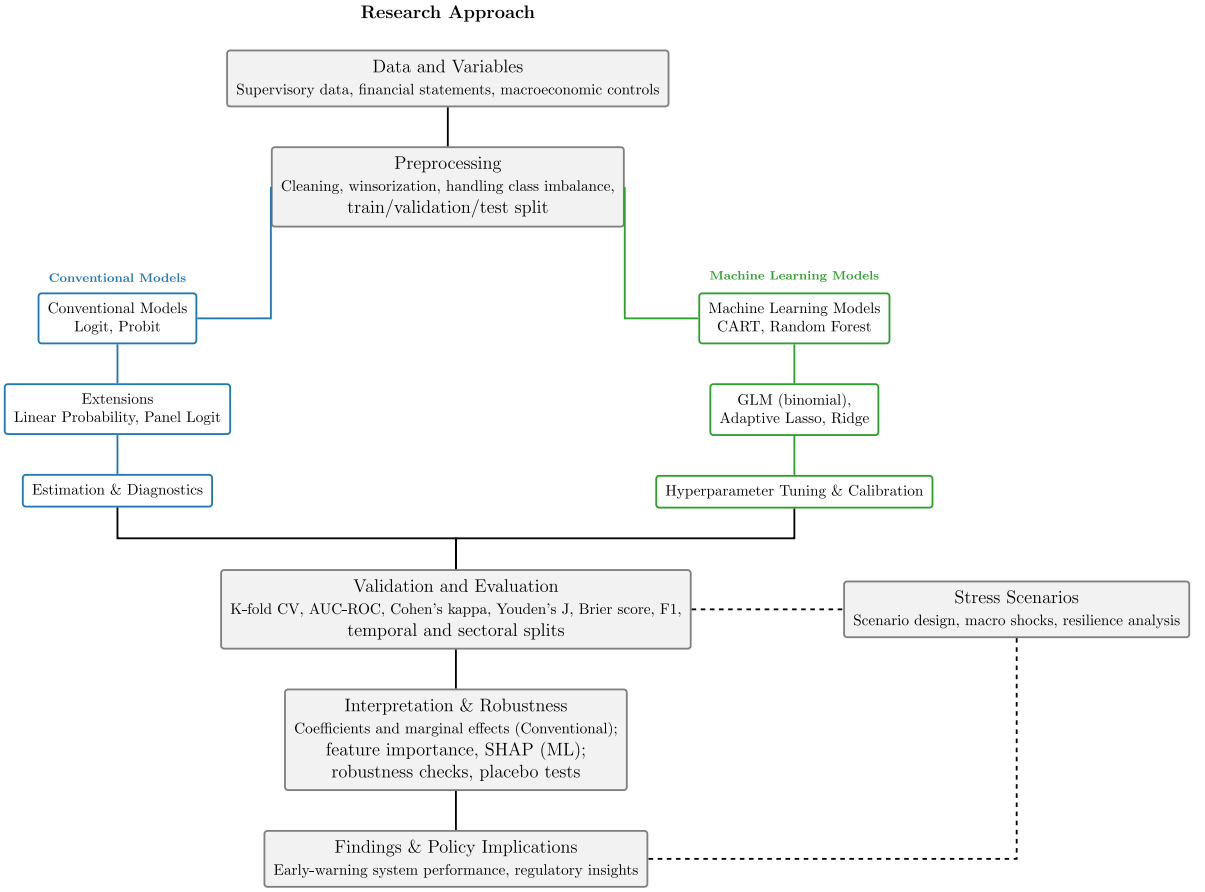


Fig. 7. End-to-end research approach for predicting Italian bank defaults. Stress scenarios are represented as a complementary branch to the main validation–interpretation–findings workflow.

Table 3
Benchmarking sample formation approaches for training and validation.

Set type	Approach	ND	D	Total = ND+D	D, % of Total
Training	(a) as is	780	14	794	1.76
	(b) over	780	120	900	13.33
	(c) over-under	723	177	900	19.66
	(d) ROSE	406	388	794	48.87
Validation		258	7	265	2.64
Training + Validation	as is	1038	21	1059	1.98

Note: ND refers to non-defaulted observations, while D signifies defaulted observations. In the context of Validation, an identical number of observations was consistently utilized for testing, regardless of the sample composition during training. This is why we present four rows for Training and a single row for Validation. For a comprehensive understanding of the four approaches (a) to (d) employed to create the initial training sets, refer to the paragraph 'ML Fitting and Sampling' from the above Section 4.1.1.

Following the presentation of these models, this section advances with a methodological discussion pertaining to factor weights, highlighting the novel contributions of this study. In cases where financial statements were missing, the procedure used to recover the data is described in Appendix B. Lastly, we provide a compilation of classification methods together with precision metrics. Real-world financial data are often subject to various forms of uncertainty (e.g., measurement errors, missing values, stochastic variability). A detailed discussion of how our framework addresses these issues, and their implications for robustness, is presented in Appendix C.

4.1.1. Machine learning (ML)

In the realm of machine learning, the analysis is segmented into two distinct steps: the first encompasses fitting and sampling, while the second entails the verification of robustness.

ML fitting and sampling. Concerning fitting and data sampling, we employed the aforementioned ML models in various scenarios: (a) utilizing imbalanced data directly, (b) oversampling data by introducing historical defaults, (c) implementing over-under sampling, which entails generating additional instances of the smaller class while simultaneously reducing non-defaulted observations, and (d) applying random over-sampling (ROSE) as proposed by Menardi and Torelli (2014). In ROSE, new artificial observations are generated through the estimation of a kernel density function.

A comprehensive comparative depiction of these four sampling approaches is provided in Table 3.

Robustness check. To establish the robustness of the outcomes, each method is subjected to a 10-fold cross-validation procedure, utilizing the common validation sample outlined in the 'Validation' row of Table 3. In this process, the algorithm randomly selects 10% of the sample as the test set, while the remaining 90% functions as the training set. Subsequently, multiple non-overlapping extractions are carried out from the 10% subset, and the model is re-run each time. The resultant outcomes, framed in terms of goodness-of-fit (classification) metrics, are then averaged over these ten predictions.

Table 3 provides a comprehensive breakdown of the samples designated for both training and validation purposes. In the overall count, the dataset comprises 1059 observations, which include the 14 outliers mentioned in Section 3.4. As a result, 1045 observations remain (=1059-14), and these are detailed in Table B.11.

Upon accounting for the one-year lag for 100 banks — equating to 100 observations — the dataset further narrows to 946 observations, representing the one-year shifted Default flag. These figures are presented in Table B.11.

4.1.2. Conventional approaches

As an alternative to the ML approach, we propose a conventional method for PD model development. We initiate this process by conducting a single-factor analysis, leveraging the availability of 72 financial indicators detailed in Table B.11.

Specifically, we focus on those that are easier to identify in financial statements because they are "historical" macro-items consistently used in financial reporting, independent of changes in accounting principles and type of bank considered. Namely we consider, ROA (PROFIT/TOTAL ASSETS), PN/TA (NET EQUITY/TOTAL ASSETS), BCC DUMMY, and CONSTANT (i.e. the intercept of the model).

For subsequent analysis, we focus on variables that exhibit statistically significant correlations with the default flag at a 5% significance level (indicated by a single star in Table B.11). Additionally, we explore the possibility of interrelatedness between chosen PD drivers, aiming to identify potential multicollinearity.

For example, if Return on Assets (ROA) is correlated with a cash flow-based indicator, we prioritize ROA. This approach ensures both model comparability with previous studies — where we employed ROA — as well as the mitigation of multicollinearity concerns.

Regarding the conventional approaches, we explore a total of eight distinct models, each detailed in Table 4. The outcomes derived from these analyses are documented in Table 5.

Drawing inspiration from Diebolt (2015), and in contrast to the ML approach of model validation, we opt to estimate model parameters employing the entire (full) data sample. This choice avoids the partitioning of data into training and testing segments. Diebolt argues that, in hindsight, one could strategically fit the best model over a subset of out-of-sample (out-of-time) historical

Table 4
Conventional Approach: Probit Model Specifications.

Model	Description
Pr01	From the entire set of variables excludes the statistically insignificant drivers in a step-wise fashion
Pr02	Starts with an empty set and step-wise new statistically significant variables are added
Pr03	Includes all variables from single-factor analysis and after exclusion of the collinear drivers
Pr04	Excludes statistically insignificant variables from Pr03 specification
Pr05	From Pr04 interactions with small bank dummy indicator (BCC)
Pr06	Specification mostly one-to-one as suggested by Moody's Analytics (2016) for comparison/benchmarking where: Return on assets (ROA) enters directly as I4ROA; Equity to assets is also available as I2PNTA; Non-performing assets is proxied by I25RRV_130TA; Change in ROA is I4ROAd; Provision to loans is proxied by I14RRV_CCVC; Loan-to-Deposit ratio comes from I17CVCDVC.
Pr07	Improvement version of Pr05 in which significant variables are taken from Pr06
Pr08	Includes all variables from specifications Pr01-Pr07

data. Furthermore, the very Bayesian updating principle inherent in ML tools mandates a full sample estimate, as underscored by Diebold (2015).

Consequently, we determine the most suitable model by assessing the statistical significance of coefficient estimates and their consistency across various specifications. Notably, we demonstrate that deviations from this consistency are a consequence of multicollinearity. This, unfortunately, hinders the accurate interpretation of individual coefficients, in contrast to scenarios like Moody's Analytics (2016).

4.1.3. Diversification indices

Following Manthoulis et al. (2021), in Appendix E we construct Herfindahl–Hirschman-type diversification indices across assets, income, deposits, and loans to capture banks' diversification patterns. Results indicate that these indices have limited short-term predictive power for default, whereas Johansen cointegration tests reveal strong long-term equilibrium relationships between diversification and key financial metrics. These findings suggest that diversification may influence bank stability over the long run, though its short-run predictive value is limited, and further research is needed to explore potential nonlinear effects, which are beyond the scope of this study.

4.1.4. Macro variables

To assess whether macrovariables can improve the *Probit* model, a dedicated analysis described in Appendix F was performed. Granger causality tests indicate that only lagged (two-period) macroeconomic indicators — namely CPI, GDP growth, and interest rates — significantly predict bank default risk, suggesting a delayed macro-financial effect. Regression results find that only lagged GDP growth is statistically significant, with higher growth reducing the probability of default; however, residual diagnostics suggest possible nonlinearity in this relationship. Interaction analyses reveal that firm profitability (ROA) remains a strong predictor of default, yet its interaction with GDP growth — whether modeled linearly or nonlinearly — does not provide additional predictive power. Furthermore, stress scenario simulations demonstrate greater resilience of cooperative banks (BCC) under adverse economic conditions. Overall, while certain macrovariables exhibit predictive power for bank default risk, their incorporation into the selected model provides limited improvement.

4.2. Factor weights and contributions

This section serves to highlight the novel contributions of our study by comparing the obtained results with those available in existing literature.

4.2.1. Factor weights and ML

Commencing with the premise that many ML-based PD models are often perceived as "black boxes", it is noteworthy that while model properties like discriminatory power may be reported (as seen in Shibitov and Mamedli (2019)), the significance of individual factors is often not disclosed. A notable exception to this norm is evident in a study by Dendramis et al. (2020, p. 39, Table 5), where authors present factorial sensitivities — commonly referred to as 'covariate effects' — equivalent to marginal effects. However, the exposure of "white box" PD models — ones that offer insights into estimated coefficients, marginal effects, and similar aspects — is a rarity. This scarcity of transparency is particularly pronounced in the banking sector. When seeking candidates for comparison, it is apparent that works such as the Moody's models for the USA in Kocagil et al. (2002), global banks in Moody's Analytics (2016), Italian banks in Ferriani et al. (2019), and Russian banks in Penikas (2022) are potential contenders. Yet, making direct comparisons among these studies proves challenging. For instance, Kocagil et al. (2002) presents factor contributions to the PD model that are consistently non-negative and collectively sum to 100%, whereas Moody's Analytics (2016) displays model weights that may encompass negative values and do not cumulatively reach 100%. Additionally, neither Kocagil et al. (2002) nor Moody's Analytics (2016) elucidate the practical methodologies employed to determine these contributions and factor weights, respectively.

4.2.2. Black box versus white box and the need for a comparison

Given that the articles by Ferriani et al. (2019) and Penikas (2022) are devoid of commercial sales motives, they both disclose details about the coefficients. This raises a quandary concerning the benchmark representation of our PD model for Italian banks in comparison to the aforementioned four 'white box' models, especially those by Moody's.

In order to facilitate a robust comparative analysis, we propose a viable approach to derive contributions and factor weights, aiming to render our model comparable to those by Kocagil et al. (2002) and Moody's Analytics (2016). We first elucidate our understanding of contributions and weights. As weights are not confined to positive values and can vary in sign, they should align with estimated coefficients and/or marginal effects. Notably, the weights in Moody's Analytics (2016) do not culminate in 100%—for instance, they aggregate to roughly 30% in the one-year PD model and −8% in the five-year PD model. Intriguingly, the absolute values of these weights collectively sum up to a precise 100%.

This prompts the question of how to delineate factor weights subsequent to estimating coefficients in nonlinear models such as logit models featured in Ferriani et al. (2019) or *Probit* models highlighted in Penikas (2022).

4.2.3. Comparison of weights and contributions of ML and conventional approaches

Let us consider $\hat{\beta}_{ik}$ as an estimated coefficient for the ith PD driver that constitutes the kth PD factor. Furthermore, let W_j represent the weight of the jth PD factor (which may encompass various PD drivers as distinct i variables), and C_j denote the contribution of the jth PD factor. We establish the requirement that contributions collectively amount to 100%, while allowing weights to signify the direction (sign) of impact. We diverge from the *Probit* functional form. Eqs. (1) and (2) reveal that the forecasted average PD for the portfolio ($\hat{PD}$) can be approximated as the value of the Gaussian cumulative distribution function N at the point $\hat{y}$. This $\hat{y}$ is a result of the linear combination of mean driver values $\bar{x}ik$ multiplied by the estimated coefficients $\hat{\beta}ik$, along with the intercept $\hat{\beta}_0$.

$$\hat{PD} \approx N(\hat{y}), \tag{1}$$

where

$$\hat{y} = \hat{\beta}_0 + \sum_{k=1}^{J}\sum_{i=1}^{I} \hat{\beta}_{ik} \cdot \bar{x}_{ik}, \tag{2}$$

where $\bar{x}_{ik}$ is the sample mean of the ith variable falling into the kth PD factor (we compute the marginal effects $\frac{dF}{dx_{ik}}$ exactly at the point of $\bar{x}_{ik}$); β_0 is the intercept.

Subsequently, we can calculate the weights and contributions in accordance with Eqs. (3) and (4), respectively. Notably, we are decomposing the value of the latent factor $\hat{y}$ rather than the average PD prediction ($\hat{PD}$). It is important to acknowledge that there might be a distinction in the marginal effect of a specific factor due to the nonlinear transformation from $\hat{y}$ to $\hat{PD}$ in the *Probit* model (which applies similarly to logit models and the designed ML models). Nonetheless, considering the undisclosed nature of Moody's method for computing factor weights and contributions, the approach detailed here emerges as the simplest, feasible, and transparent solution.

$$W_j = \frac{\sum_{i=1}^{i=I}(\hat{\beta}_{ik} \cdot \bar{x}_{ik})|_{k=j}}{\sum_{k=1}^{k=J}\left(\sum_{i=1}^{i=I}(abs(\hat{\beta}_{ik} \cdot \bar{x}_{ik}|_{k=j}))\right)}. \tag{3}$$

$$C_j = \frac{\sum_{i=1}^{i=I}(abs(\hat{\beta}_{ik} \cdot \bar{x}_{ik})|_{k=j})}{\sum_{k=1}^{k=J}\left(\sum_{i=1}^{i=I}(abs(\hat{\beta}_{ik} \cdot \bar{x}_{ik}|_{k=j}))\right)}. \tag{4}$$

Indeed, Eq. (4) demonstrates that the denominator for the contribution of the PD factor (and also its weight as per Eq. (3)) might diverge from the value of $\hat{y}$ in cases where at least one mean driver value or a coefficient estimate is negative. Employing the proposed method for weight calculation helps circumvent the issue of compensation in factor contributions. To illustrate this, consider a simple PD model comprising two factors. Suppose these factors possess equal sample means of 1 and estimated coefficients of +1 and −1 respectively. Consequently, the point estimate for the overall PD factor will be zero ($\hat{y} = (1\cdot+1)+(1\cdot-1) = 0$), leading to a PD forecast of 50% assuming a Gaussian CDF. At this juncture, the individual contributions for each factor cannot be determined. For instance, when dividing a coefficient estimate by the zero-sum of $\hat{y}$, we obtain values of infinity, which is not interpretable. However, the approach we suggest yields factor contributions of 50% and 50%, while the factor weights in accordance with Eq. (3) are +50% and −50%, making the interpretation more meaningful.

The approach outlined in Eq. (3) and (4) offers the advantage of comparability with previous studies. However, in this comparison, we are somewhat limited by the work of Ferriani et al. (2019) since they provide the estimated coefficients $\hat{\beta}_{ik}$ without the corresponding sample means $\bar{x}_{ik}$. In instances where $\bar{x}_{ik}$ is not specified, we assume $\bar{x}_{ik} = 1$. While Ferriani et al. (2019) does offer the calibration of the model across four specifications for PD prediction horizons of one year (4 quarters) and one and a half years (6 quarters), they do not clarify which of the four specifications per horizon should be considered as the most appropriate. Consequently, we calculate the mean coefficient estimates across the four specifications and focus on the one-year prediction horizon for the purpose of comparability with the PD models developed for US banks in Kocagil et al. (2002), world banks in Moody's Analytics (2016), and Russian banks in Penikas (2022).

4.3. Classification and precision metrics

To compare the results obtained from the models, a number of accuracy metrics are employed. Namely: (a) accuracy, (i.e. the proportion of real detection in a class); (b) sensitivity (i.e. the rate of true positive predictions (TPR)), (c) p-value for testing the null of accuracy exceeding the No-Information Rate (NIR) from Kuhn (2008, p. 15), (d) area under the Receiver Operating Curve (AUC, AUROC), (e) kappa of Cohen (1960), (f) J of Youden (1950, p. 33). These metrics collectively provide insights into the performance of the models in terms of their predictive accuracy and ability to distinguish between defaulted and non-defaulted instances.

5. Findings

This section presents the main empirical findings. We begin by using LASSO feature selection on two datasets — DS(B73) and DS(B100) — to identify the most material PD drivers (notably ROA and net credit-risk adjustments, I68.RRV130.CVC). We then compare conventional econometric models (logit, probit, linear probability, and panel logit) with machine-learning classifiers (CART, random forest, and GAM). On the larger DS(B100) sample, GLM specifications perform best overall: the GLM+ROSE configuration attains Youden's $J = 0.692$ with AUC = 0.846, and outperforms CART and random forest, while smaller-sample results reveal mixed performance and signs of overfitting. To keep the narrative concise, we discuss only the most informative coefficients and headline metrics (AUC and J) in the text; the full regression tables and ROC analyses are reported in Appendix D.

5.1. Machine learning

The LASSO method applied first at the smaller sample DS(B73) and then at the larger sample DS(B100), suggests that there are two most material PD drivers:

- ROA = Profit / Total Assets (I4.ROA);
- Net impairment (losses) reversals on (a) financial assets measured at amortised cost (b) financial assets measured at fair value through other comprehensive income (ex IAS 39 Item 130) / Credits Vs Customers (I68.RRV130.CVC).

These indicators are directly comparable to those used in prior studies and both indicators are related to profitability and capitalization. The visual analysis presented in Fig. 6 highlights the significance of profitability and capital base as key PD drivers for Italian banks. In conclusion, the utilization of the smaller sample consisting of 73 banks without consolidated reporting [DS (B73)] did not yield an optimal model. This is because, even though the median AUC of GLM appears to be higher than that of machine learning models, the results are generally mixed. In some cases, the AUC of the latter is close to 0.5 (indicating no discriminatory ability), while in other cases, the AUC is very close to 1 (indicating overfitting).

In contrast, the larger dataset of 100 banks favors the conventional GLM. The best performance with the machine learning approach is achieved by the GLM regression ROSE algorithm, which yields a Youden's J value of 0.692 and an AUC of 0.846. The next best model is also the GLM regression, using the original Both data approach, with a Youden's J value of 0.558 and an AUC of 0.779, details available in Tables D.12. As a comparison, the model utilized by Calabrese and Giudici (2015), featuring the generalized extreme value model and the logistic regression, on a sample of 783 Italian banks (1996–2011) provides AUC values of 0.7134 and 0.7889, respectively. More details can be found in Tables D.13–D.18. As a result, except for the cart model with ROSE approach with an AUC of 0.819 and a Youden's J value of 0.638 and those models with an AUC that seem influenced by overfitting by comparing the median of the AUC of the several approaches applied, it seems that the GLM scored higher which seems to suggest more robust results. In this way we have demonstrated that the conventional model outperforms both the random forest and cart models. To further validate this, we conduct a robustness check using the *Probit* model within the conventional configuration.

5.2. Conventional approach

Table 5 provides a summary of coefficient estimates for the models listed in Table 4. According to the results, the best model is Pr05 because it has only statistically significant factors, fewer regressors, provides a pseudo R-squared value similar to alternative models, one of the highest AUCs, and is not prone to multicollinearity (no inflation in the estimated coefficients).

First, it is worth noting that similar to Ferriani et al. (2019), the dummy variable representing small cooperative banks (BCCs) is statistically significant, even though we find a negative coefficient, whereas Ferriani et al. (2019) reports a positive coefficient. This discrepancy might be anticipated since the paper by Ferriani et al. (2019) suggests giving more attention to the less significant banks. However, our dataset encompasses not only smaller banks but also the largest banks in Italy. Moreover, it is quite plausible that the 2016 reform of cooperative banks had a positive impact on the sustainability of cooperative banks in Italy. This is supported by the fact that the BCC dummy had a positive sign until 2016 in Ferriani et al. (2019), even though it exhibits a negative effect in our case until 2020. This difference could be attributed to a delay effect or the favorable impact of the BCC reform that we will delve into further in Section 6.2.

Second, we discover that half of the variables utilized in Moody's Analytics (2016) are not significant for Italian banks, including factors like the loan/deposit ratio, assets, or credit quality. This implies that the Moody's Analytics (2016) model cannot be directly applied to Italian banks without careful consideration.

Third, we observe that the 'Change in ROA' variable, introduced in Moody's Analytics (2016), demonstrates statistical significance across specifications Pr01, Pr02, Pr03, Pr04, Pr06, and Pr07.

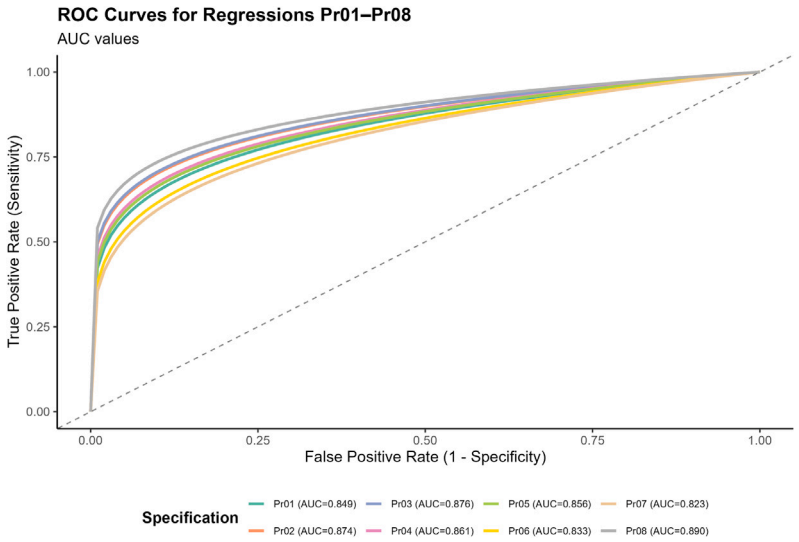


Fig. 8. ROC curves for alternative regression specifications (Pr01–Pr08).

Fourth, our results are in agreement with the pre-bank reform study (Calabrese and Giudici, 2015), where asset quality (i.e. Loan Loss Reserve and Loans to Assets), return on assets (ROA), solvency (Equity/Total Assets), liquidity (Liquid Assets/Total Deposits), and capital ratios (Tier 1 Capital Ratio) are relevant for estimating the PD.

As we turn our attention to the remaining Pr05 specification, it becomes necessary to closely examine the corresponding coefficient estimate and conduct a comparative analysis of the results. On one hand, the coefficient estimate for 'Change in ROA' is unstable, exhibiting variations from 22 in Pr03 to 40 in Pr02, equating to roughly a twofold difference (or 200%). This variation stands in stark contrast to the stability of the I2PNTA coefficient estimate, which varies only from 7.1 in Pr08 to 9.3 in Pr04, reflecting a 30% change. On the other hand, the inclusion of 'Change in ROA' leads to alterations in the 'ROA' coefficient estimates, shifting from −59 in Pr05 to −77 in Pr07. This situation reflects a classic example of multicollinearity. In conclusion, using both 'ROA' and 'Change in ROA' factors results in relatively similar overall PD predictions (see Fig. 8).

When interpreting the coefficients separately, there is a misleading indication of a higher contribution from 'ROA'. Due to this reason, and according to Table 5, we consider the Pr05 specification to be the most suitable, i.e.

$$Pr(D = 1) = N\left(-1.261 - 7.556 \cdot LR - 59.484 \cdot ROA - 0.568 \cdot BCC\right), \tag{5}$$

where D corresponds to the flag of the of default; N is the Normal (Gaussian) CDF; LR is the leverage ratio (the ratio of total capital to assets); ROA indicates return on assets; BCC is the indicator whether the bank is a small cooperative or not.[1]

At this point, we are able to compare our findings with the following models:

- Italy (Pr05) - our best model according to Eq. (5);
- Italy (Ferriani et al. 2019) - model from Ferriani et al. (2019);
- Italy (Moody's 2016) - model from specification Pr06 in Table 5. It has the very same PD drivers as mentioned in Moody's Analytics (2016), but recalibrated on our data;
- Russia (2022) - model from Penikas (2022) on Russian banks;
- World (2016, Moody's) - model weights here are taken precisely from Moody's Analytics (2016) that is based on a sample of 90 countries around the world;
- USA (2002, Moody's) reflects the factor contributions from Kocagil et al. (2002) on American banks.

The comparison of the PD factor weights for the first five specifications aforementioned is presented in Fig. 9, while the benchmarking of the PD factor contributions for the entire set of six specifications is shown in Fig. 10.

As we can observe from Fig. 9, the model weights predominantly indicate the same impact direction. Higher profitability (ROA), along with the associated capital base (Equity to Assets), has a negative influence on PD. At the same time, noticeable distinctions emerge in the PD drivers for Italian banks. The capital base holds much greater significance for Italian banks compared to their counterparts worldwide. Moreover, the considerable reduction in PD among Italian banks originates from other factors, namely the intercept and the BCC dummy. In contrast to Ferriani et al. (2019) and Penikas (2022), we do not uncover statistical evidence

[1] Note that in Eq. (5) LR corresponds to I2PNTA (i.e. I2 PN/TA = Net Equity/Total Assets), ROA is I4ROA (i.e. I4 ROA = Profit/Total Assets), and BCC corresponds to the dummy for small banks (Banca di Credito Cooperativo, BCC).

Table 5
Higher R-squared is often driven by multicollinearity and exorbitant scales of estimated coefficients.

Variable	Pr01	Pr02	Pr03	Pr04	Pr05	Pr06	Pr07	Pr08
_cons cons = constant (intercept)	−1.945	−1.562***	−0.469	−1.126***	−1.261***	−1.639***	−1.341***	1.337
I1ROE I1 ROE = Roe = Profit/Net Equity	−29.940***							−13.666
I2PNTA I2 PN/TA = Net Equity/Total Assets			−8.797*	−9.328*	−7.556*	−8.201*	−8.508**	−7.119
I2PNTA_BCC I2 PN/TA BCC = (Net Equity or Equity)/Total Assets of BCC				5.317				13.169
I4ROA I4 ROA = ROA = Profit/Total Assets		−243.758***	−103.830***	−57.223***	**−59.484****	−79.499***	−77.413***	−68.364
I4ROA_BCC I4 ROA BCC = ROA = Profit/Total Assets of BCC				−6.983				42.706
I4ROAd I4 ROAd = Change in ROA: ROA(t) - ROA(t-1)	31.543**	40.077***	22.928*			28.737**	28.361**	37.866**
I6PTA I6 P/TA = Loans/Total Assets			1.278					−0.488
I8MIMID I8 MI/MID = Interest Margin/Brokerage Margin			−0.329					−1.058
I14RRV_CCVC I14 RRV C/CVC = Net adjustments Write-backs for deterioration of Loans/Loans						−3.348		21.753
I16CCNPN I16 CCN/PN= Net Working Capital/Net Equity			−0.155					−0.118
I17CVCDVC I17 CVC/DVC= Receivables from customers/Deposits from customers						0.254		0.357
I25RRV_130TA I25 RRV 130/TA = Net adjustments write-backs for impairment of loans/Total Assets						3.845		−3.415
I28RVATA I28 RVA/TA=Value adjustments of goodwill/Total Assets			102.192*					63.582
I36RAPSIPNT I36 RAPSI/PNT=Result Before Extraordinary Income and Taxes/Tangible Net Equity								3.173
I37RAITA I37 RAI/TA=Pre-Tax ResultTotal Assets		294.814***						−17.117
I40RAPSIPN I40 RAPSI/PN=Result Before Extraordinary Income and Taxes/Net Equity								−13.314**
I41RAIPN I41 RAI/PN=Pre-Tax Result/Net Equity	8.450***							21.825
I44RAPSIRG I44 RAPSI/RG=Result Before Extraordinary Income/Operating Result			0.012					0.019
I48MIDSA I48 MID/SA=Brokerage/Administrative Expenses			−0.448					0.026
I51BKS_LNTA I51 BKS LN(TA)=Bank Size (Proxy Variable) = Ln (TA)			−0.102*					−0.117*
I60AFTAPN I60 AF/(TA-PN)=Financial assets/(TA-PN)		−2.840***						−2.798*
I68RRV130CVC I68 RRV130/CVC=Adjustments Rep. A Val. Net for Det. (V.130))/Credits Vs Customers			9.857					−17.853
BCC BCC=Dummy for small bank (Banca di Credito Cooperativo, BCC)	−0.727**		−0.949***	−1.012	**−0.568****			−1.792*
N	849	849	849	946	**946**	849	849	849
r2_p	0.291	0.307	0.316	0.255	**0.253**	0.24	0.236	0.403
AUC	0.8493	0.8742	0.8765	0.8610	**0.8560**	0.8330	0.8226	0.8900
Number of Terms	5	5	13	6	**4**	7	4	24

*** Significant at 1% ; ** - at 5%; * - at 10%.
_cons - intercept; N - number of observations, r2_p - pseudo R-squared; AUC - area under ROC curve.
Pr05 (in bold) is the best specification because it has only statistically significant factors, fewer regressors, provides a pseudo R-squared value similar to alternative models, one of the highest AUCs, and is not prone to multicollinearity (no inflation in the estimated coefficients).
Legenda: Long description of variables. _cons = constant (intercept); I1 ROE = Profit (Loss) for the year/Equity; I2 PN/TA = Equity/Total Assets; I2_PN/TA_BCC = Net Equity/Total Assets = Equity/Total Assets of BCC; I4 ROA = Profit (Loss) for the year/Total Assets; I4_ROA_BCC = Profit (Loss) for the year/Total Assets of BCC; I4_ROAd = Change in ROA: ROA(t) - ROA(t-1); I6 P/TA = Financial assets measured at amortised cost: (b) Loans to customers /Total Assets; I8 MI/MID = Net interest margin/Operating income; I14 RRV C/CVC = Net value adjustments for credit risk (a) financial assets measured at amortised cost /Financial assets measured at amortised cost : (b) Loans to customers ; I16 CCN/PN = Net Working Capital/Equity; I17 CVC/DVC = Financial assets measured at amortised cost: (b) Loans to customers Financial liabilities measured at amortised cost: (b) Deposits from customers ; I25 RRV 130/TA = Net value adjustments for credit risk of (a) financial assets measured at amortised cost (b) financial assets measured at fair value through other comprehensive income/Total Assets; I28 RVA/TA = Net value adjustments write-backs on intangible assets (goodwill)/Total Assets ; I36 RAPSI/PNT = [Profit (Loss) from continuing operations before tax - Profit (loss) from equity investments] Tangible Equity; I37 RAI/TA = Profit (Loss) from continuing operations before tax/Total Assets; I40 RAPSI/PN = [Profit (Loss) from continuing operations before tax - Profit (loss) from equity investments]/Equity; I41 RAI/PN = Profit (Loss) Before Tax/Equity; I44 RAPSI/RG = Profit (Loss) From Continuing Operations/Net Operating Profit (Loss); I48 MID/SA = Operating income/Administrative expenses; I51_BKS_LN(TA) & Bank Size (Proxy Variable) = Ln (TA); I60 AF/(TA-PN) = (Financial assets measured at fair value through profit or loss +Financial assets measured at fair value through other comprehensive income)/(Total Assets - Equity) or (Total Liabilities); I68 RRV130/CVC = Net value adjustments for credit risk of (a) financial assets measured at amortised cost (b) financial assets measured at fair value through other comprehensive income/Financial assets measured at amortised cost: (b) Loans to customers.

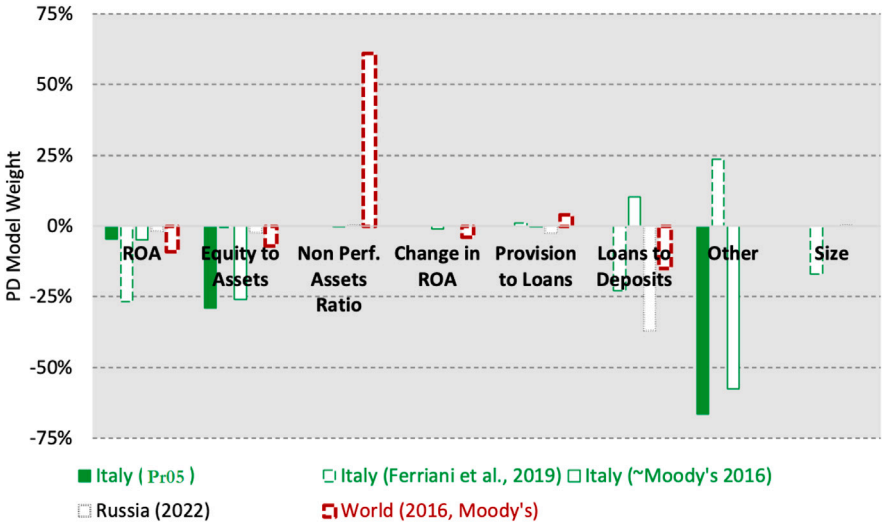


Fig. 9. The PD of Italian banks is primarily influenced by their relative earnings (ROA and changes therein), with relatively less sensitivity to asset and loan book quality compared to banks in the US and Russia.

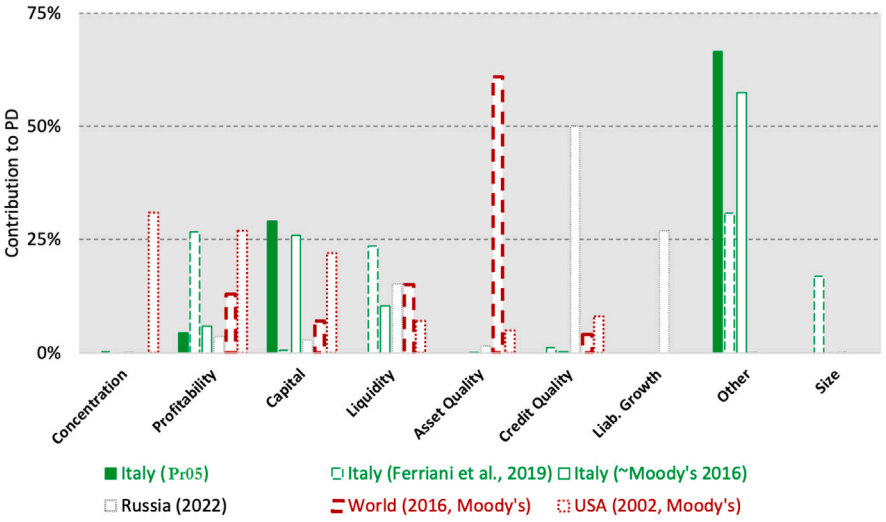


Fig. 10. Italian banks are equally sensitive to leverage (equity to assets), as the US and Russian banks.

supporting the notion that a bank's size plays a crucial role in predicting Italian idiosyncratic PD. It could be argued that, to some extent, our findings align with Kocagil et al. (2002) and Moody's Analytics (2016). However, we remain uncertain whether Moody's RiskCalc model dismissed the size factor as insignificant or did not consider it at all, resulting in a model exclusively reliant on relative ratios.

In terms of the absolute contribution of the PD factor, we can also draw comparisons between our findings and those of Kocagil et al. (2002) concerning US banks, despite the fact that their study dates back mainly two decades. As depicted in Fig. 10, it becomes apparent that liquidity could be a relatively common PD factor, contributing approximately 15%–20% to the PD forecast for Italian, Russian, US, and other banks. However, it is worth noting that we did not find liquidity to be as significant in our best model.

The key conclusion is that Italian banks possess unique idiosyncrasies that must be taken into account when formulating a PD model. The Moody's Analytics (2016) model, designed for global banks, bears the closest resemblance to our model. However, it incorporates redundant factors that hold no significance for Italian banks (such as Change in ROA and NPL). Concurrently, vital factors carrying up to 50% weight in a PD model are absent. A systematic factor is captured by the intercept and the Italian-specific nature of small banks, as indicated by the BCC indicator. This underscores that previous models, when extended to Italian banks, yield misleading PD predictions. The PD model developed herein presents notable advantages for both future PD forecasts and retrospective analyses. The latter will be discussed in the upcoming Section 6.1.

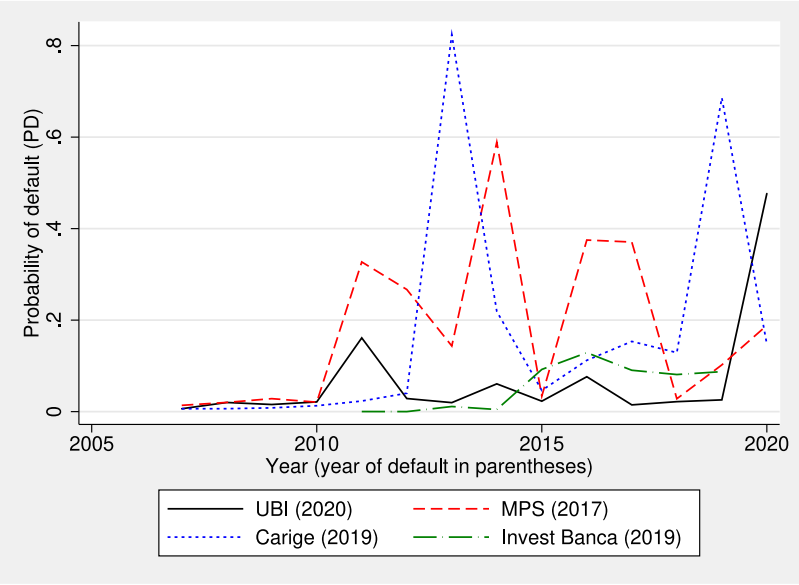


Fig. 11. Developed PD model allows to trace back the roots when the default causes emerged.

6. Discussion and policy implications

Based on the PD model developed for Italian banks, two policy implications emerge, pertaining to the supervisory approach and the effects of the 2016 BCC reform.

6.1. Modern economic history of Italian banks

As previously noted in Alonso and Carbo (2020), the advent of Basel II and the Internal-Ratings-Based (IRB) approach prompted the need for PD model development. Nevertheless, a well-developed PD model can also yield insightful historical information, as outlined below.

We have constructed a PD model based on established information regarding specific bank defaults. Once the PD model was estimated, we extended its application beyond just future predictions and current IRB capital requirements calculations. We also conducted retrospective analysis, enabling the prediction of PD values for any bank, even those that have already defaulted, as illustrated in Fig. 11.

Supervisors can gain insight into the historical performance of the banking sector and ascertain whether certain supervisory measures could have been implemented earlier. To illustrate, consider the case of the defaulted Carige bank in 2019 (indicated by the dotted blue line in Fig. 11). Notably, the PD in the year of insolvency approached 70%. However, a remarkable observation emerges: the retrospective PD forecast exceeded 80% in 2013. This suggests that had stricter supervisory measures been imposed on Carige in 2013, it might have averted default in 2019 and continued to operate successfully.

A comparable analysis can be executed for what are commonly referred to as healthy banks, as illustrated in Fig. 12. The prior mention of the relevant Low Default Portfolio (LDP) term pertains to Italian banks as a collective segment. One of the challenges inherent to an LDP segment involves PD estimation in cases where no defaults have been recorded within the segment. In this context, the question arises regarding the PD applicable to banks that have never experienced defaults. This is applicable to significant Italian banks such as Intesa, UniCredit, Iccrea, and others.

The developed PD model provides a means to address this inquiry by facilitating a relative ranking of non-defaulting banks and offering an estimate of potential PD values. For instance, in the case of Iccrea, which has no history of defaults, its PD registers as the highest in 2020 and demonstrates a minor upward trend (as illustrated by the blue dotted line in Fig. 12). Similarly, UniCredit's condition appears robust, with a projected PD of approximately 5% in 2020 (as indicated by the dashed red line in Fig. 12). Notably, during 2013 and 2016, the bank encountered challenging periods, as suggested by the model, with the PD surging above 20%. Conversely, Intesa outperforms UniCredit, despite both experiencing a similar up to 15% rise in PD in 2011 (as depicted by the solid black line in Fig. 12). However, Intesa's PD has exhibited a consistent decline, whereas UniCredit's PD has demonstrated greater volatility.

A concluding observation pertains to Banca di Asti, which emerges as the most stable among all those under consideration. In this instance, the lowest PD projection hovers around 2%, maintaining notable stability over the past decade (as depicted by the long dashed green line in Fig. 12). This stability endures irrespective of defaults among less healthy banks and PD peaks for healthier ones.

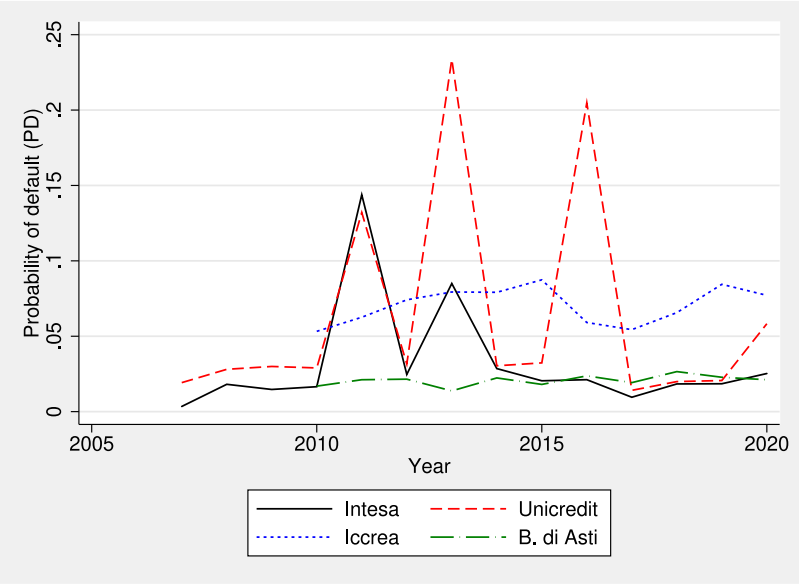


Fig. 12. Developed PD model enables to differentiate credit risk with respect to liabilities of non-defaulted banks.

Table 6
Difference-in-Differences Matrix for BCC Reform in Italy.

BCC Dummy	Indicator	Before-After Dummy		Total
		Before 2016	After 2016	
non-BCC	N	334	208	542
	DR	.0242	.0745	.0407
BCC	N	278	225	503
	DR	.0109	.0000	.0066
Total	N	612	433	1045
	DR	.01818	.0352	.0243

Note: N = number of observations; DR = default rate (share of total N).

6.2. 2016 BCC reform

The 2016 reform of small cooperative (BCC) banks in Italy, as highlighted in EACB (2016), has been previously mentioned. In line with the approach presented by Ferriani et al. (2019), we introduced a BCC dummy variable. However, astute readers might question why we did not conduct a treatment effect estimation for the BCC reform. A common methodology to address this is by employing a difference-in-differences (DiD) framework. For example, recent research such as Duong (2021) utilizes DiD analysis to demonstrate that inflation targeting (IT) does not necessarily enhance the well-being of emerging economies. Despite lower inflation rates observed in IT-adopting countries during pandemics, the associated disinflation came at a considerable cost.

Efforts were made to enhance the aforementioned Pr05 specification outlined in Eq. (5). This improvement involved the inclusion of a before-after $Y2016$ dummy variable and its interaction counterpart, $BCC_2016 = BCC \cdot Y2016$. However, the model excluded the interaction dummy, which was the central focus, due to its perfect prediction of non-default cases. The distribution of observations within the DiD matrix is reasonably even, as depicted in Table 6. Nonetheless, a noticeable disparity emerges in the default rate (DR) for BCC banks after the year 2016. It dropped from 1% in the 'Before' 2016 period to zero. Conversely, non-BCC banks in Italy exhibit a contrasting trend. The DR for non-BCC banks increased from 2.4% in the 'Before' 2016 period to 7.4% in the post-reform period.

Thus, the BCC 2016 treatment effect on the DR is the difference of differences in DR equaling to −6 pp. as follows:

$$DR_{A-B}^{BCC} - DR_{A-B}^{Non-BCC} = \left(0.0\% - 1.1\%\right) - \left(7.4\% - 2.4\%\right) = -1.1\% - 5.0\% = -6.1\%$$

where $A - B$ stands for the 'After' minus 'Before' difference.

We acknowledge the limitations of our approach, particularly our inability to test the parallel trend assumption, a step taken in studies such as Mäkinen (2021, p. 15–16) or discussed in Duong (2021, p. 9). Notwithstanding these recognized limitations, our approach provides a second-best estimation, leading us to the conclusion that the BCC reform had a positive impact on the likelihood of default for small cooperative (BCC) banks in Italy. In other words, BCC banks demonstrated reduced susceptibility to default following the 2016–2019 BCC reform. Employing basic cross-validation, we find support for the notion that BCC banks were

relatively vulnerable before the reform, as discussed in Federazione delle Cooperative Raiffeisen (2015). This vulnerability likely contributed to the proposal of the reform itself.

Consistent with this evidence, Fiordelisi and Mare (2013) analyze the contribution of efficiency to the probability of default among Italian cooperative banks, showing that while these banks play a crucial role in the European banking industry, they faced significantly higher default rates than commercial banks during periods of financial stability. Improvements in cost, revenue, and profit efficiency substantially increase the survival time of cooperative banks, reinforcing the link between managerial quality and financial resilience. Similarly, Barra and Ruggiero (2022) highlight that capitalization, credit volume, and intermediation costs are major determinants of non-performing loans, and that cooperative banks have adopted more prudent credit policies after the financial crisis and the Basel reforms. Although this study focuses on a single-country sample, Italy's systemic importance and the diversity of its banking sector reinforce the broader relevance of our findings. The Italian banking system, particularly its cooperative sector, has historically exhibited distinctive patterns of credit risk and efficiency challenges, especially during periods of financial instability. Furthermore, the 2016 BCC reform introduced a globally unique dual-layer governance model that balances centralized supervision with local autonomy without creating holding companies—a structure not replicated elsewhere. This institutional uniqueness, combined with Italy's critical role within the European banking system, offers a valuable and insightful setting for analyzing bank default risk and resilience dynamics (European Banking Authority (EBA), 2016; International Monetary Fund (IMF), 2020).

Despite the absence of a formal BCC group prior to 2016, the corresponding default rate of 1.1% for BCC banks from Table 6 equates to a BB- credit rating according to the FitchRatings (2021, p. 11) Master Scale. Notably, the recent credit rating for the Iccrea group stands at BB- with a positive outlook, as indicated in FitchRatings (2022a). This suggests that the external credit agency has not observed a deterioration in the credit rating of the BCC group from 2016 to 2022.

Initially, the presented evidence showcasing the improvement in default rates (DR) for BCC banks might appear to be an additional point in favor of the EU Single Supervisory Mechanism (SSM). This evidence aligns with prior studies that have championed the merits of introducing the SSM, as demonstrated by works such as Loipersberger (2018) and Tziogkidis et al. (2020). However, delving deeper reveals a more intricate picture than what a cursory look might suggest. The situation could be more complex than it appears at first glance due to a range of factors. The immediate impact of enhanced supervisory efforts might not have been instantly evident, particularly when considering the intricate dynamics of dual-controlled BCC groups, as previously described. Simultaneously, a significant development that transpired after the establishment of BCC groups was the availability of mutual guarantees. To be precise, this support mechanism gained more prominence in 2017 when two distinct BCC groups were separated. This newly accessible BCC intra-group support likely played a pivotal role in reducing the risk profile of BCC banks following the reform. This observation is not intended to critique the SSM in a manner akin to Carboni et al. (2017). However, from our perspective, it appears that the initiative undertaken by the Bank of Italy played a more substantial role in enhancing the creditworthiness of BCC entities compared to the additional contribution from the broader pan-European SSM framework.

6.3. Conclusions

Our study has not only highlighted the superiority of the new PD model designed for Italian banks, surpassing the performance of previous banking PD models like those proposed by Kocagil et al. (2002), Moody's Analytics (2016), Ferriani et al. (2019), and Penikas (2022) when applied to the Italian banking landscape but also showcased the practical application of the developed PD model in analyzing the Italian banking system. Notably, our findings have confirmed the positive impact of the BCC reform in Italy, as evidenced by a reduction in the BCC default rate. As matter of fact, in 2023, the mutual banks boasted a higher average capital strength than the banking sector, with the CET1 of cooperative credit banks averaging 21.64%, compared to the banking sector's 14.8%. Additionally, the Tier 1 ratio of the mutual banks stood at 21.77%, surpassing the banking system's average of 16.1% (Teleborsa, 2023). This outcome further serves to validate the effectiveness of the Single Supervisory Mechanism (SSM).

Looking ahead, future research could extend the model to broader European datasets, particularly to explore whether similar governance reforms in cooperative banking sectors across other countries yield comparable impacts on default risk. Herfindahl–Hirschman-type diversification indices — constructed across key balance sheet components such as income, deposits, and loans — can be further explored to assess medium- to long-term effects and potential nonlinearities. Additionally, incorporating macroeconomic variables could offer deeper insights into the structural drivers of bank risk.

CRediT authorship contribution statement

Gianpiero Chironna: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Giuseppe Orlando:** Writing – review & editing, Writing – original draft, Visualization, Software, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

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Appendix A. Summary of the literature on bank default estimation

This section reviews the main contributions to the literature on bank default estimation, with particular emphasis on the period 2020–2025, when the prevailing regulatory and accounting frameworks — Basel III, IFRS 9, and the Italian measures on cooperative banks (BCC reform) and non-performing loans (GACS) — were already in place. Earlier contributions, particularly those published before 2015, are retained primarily as methodological or theoretical benchmarks, helping to trace the evolution of techniques from traditional statistical models to modern machine-learning approaches.

Table A.7 provides a broad international overview, situating studies in relation to their sample windows, reforms, methodological choices, and variable sets. This table highlights both the global scope of empirical analyses and the ways in which cross-country experiences have informed the assessment of bank fragility and default risk.

Table A.8 then narrows the focus to Italy, offering a structured comparison of four Probability of Default (PD) studies conducted in the Italian context: Calabrese and Giudici (2015), Ferriani et al. (2019), Chironna and Orlando (2023) and the present paper. These studies share an institutional setting but differ in terms of data frequency, distress definitions, treatment of state aid, and methodological frameworks. Presenting them side by side makes it possible to assess how the Italian empirical tradition has evolved, how successive regulatory changes have been incorporated, and how the scope of variables and modeling techniques has expanded over time.

Appendix B. Description of the sample

The source of information used in this paper comes from the public financial statements reported on the institutional website of each bank. In the case of banks absorbed by third-party banks, or which have merged, it was possible to recover the historical financial statements from the Financial Statements and Reports website in which all the previous historical financial statements are present. Where the financial statements were not present, we contacted by email the third bank directly to request the historical financial statements. In some cases where the bank changed its name, the historical financial statements were reported either on the old website or on the new website. For each bank and in particular for each financial statement, we keep a record indicating whether the source of the extracted data comes from the bank's institutional website or from an internal office of the bank.

The sample construction considered three primary factors: geographical area of expertise, corporate structure (legal form), and the accessibility of public data.

- **A - Geographical Area** The decision to employ diverse selection criteria stemmed from the limitations inherent in relying solely on one factor. If we had chosen to exclusively utilize geographical aspects for area sampling, we would have needed to consider additional factors:
  - (i) In the years following 2010, numerous merger operations caused changes in the list of banks within geographical areas, resulting in a relative reduction in the sample proportions for these areas. How should we have addressed this change?
  - (ii) Many banks operate with branches spanning multiple geographical regions, including cases where they serve the entire national territory, despite being classified based on the location of their general management or registered office (e.g., MPS, Unicredit, BPER, Intesa Sanpaolo, etc.). How should we handle such scenarios?
  - (iii) Certain geographical regions, particularly the South and the Islands, have fewer banks (e.g., in 2010, South: 105, Islands: 40) compared to other regions (North-west: 214, North-east: 241, Center: 160) Table B.9. Consequently, in regions with smaller populations, we would anticipate a greater variability in the studied phenomenon. This issue could only be mitigated by employing a higher sampling fraction in these regions compared to others, which is the approach we adopted.
- **B - Legal Form** If we had solely relied on the corporate form aspect (e.g., spa banks, cooperative banks, mutual banks, rural banks, etc.) for sample selection, we would have ensured the representation of both small, medium, and large banks without considering the geographical distribution of these banks. Although certain legal forms, like Cooperative Banks (Banche Popolari), constituted the least numerous category (with only 37 in 2010) compared to other forms (e.g., Spa Banks: 233, Bcc: 415, foreign bank branches: 75) Table B.10, this approach still presents challenges. In this scenario, too, business models with fewer representatives would naturally exhibit greater variability in the studied phenomenon. This challenge could only be mitigated by employing a higher sampling fraction compared to other forms.
- **C -Availability of Public Data from 2010 to 2020** The selection criteria for the bank sample considered not only geographical location and legal structure but also the availability of publicly accessible data spanning from 2010 to 2020. In cases where financial statements were missing, we made requests to the bank's general management to retrieve them. Banks that provided the missing financial statements were included in the sample (e.g., Banca di Bologna, Bcc Valdostana, Bcc del Veneziano, etc.). Conversely, banks that were contacted but did not provide the required data were excluded from the sample (e.g., Bcc di Brescia, Banca di Piacenza, Banca Popolare di Cividale, etc.). Challenges arose in recovering data related to branches of foreign banks, as complete data availability from 2010 to 2020 was not consistent. Retrieving missing data for this category of institutions proved to be particularly difficult, resulting in a smaller proportion of observations sampled for these entities. In light of the above, the sample was created by considering (i) geographical region, (ii) legal structure, and (iii) the availability of public data from 2010 to 2020. It is worth noting that a higher sampling fraction was applied to certain geographical areas (South and Islands) and certain legal forms (cooperative banks) to mitigate the potential impact of having fewer observations, which could introduce greater variability into the studied phenomenon.

Table A.7
Summary of the literature on bank default estimation (foregrounding 2020–2025 studies and the prevailing Basel III/IFRS 9/GACS/BCC-reform environment).

Paper	Year	Sample Window	Span (years, inclusive)	Reforms and Measures	Number of Banks in the Sample	Types of Banks	Geographic Area of the Sample	Methodology	Number of Variables Analyzed
Predicting Bank Defaults in Italy (present paper)	2025	2010–2020	10	Bcc (2016), GACS/NPL (2016), Basel III (2013), IFRS 9 (2018), Cura Italia (Mar. 2020)	100	Less Significant Institutions (LSIs; Bcc), Cooperative Banks (Popolari), Spa, Foreign Branches, Listed and Global Systemically Important Institutions (G-SII)	Italy	Logit and Probit Models; Linear Probability Model; Panel Logit; Machine Learning (CART, Random Forest); Logit with Diversification Indices; Logit with Interaction Term; Generalized Additive Model (GAM); Stress Testing and Scenario Analysis	72+ micro, 3 macro (+ lags), 4 Diversification Indices, 1 Interaction Term
Chironna and Orlando (2023)	2023	2010–2020	10	Bcc (2016), GACS/NPL (2016), Basel III (2013), IFRS 9 (2018), Cura Italia (Mar. 2020)	100	Less Significant Institutions (Bcc), Cooperative Banks (Popolari), Spa, Foreign Branches, Listed and G-SII	Italy	Machine Learning (GLM, CART, Random Forest); Conventional Approach (Logit Model)	72+
Penikas (2022)	2022	2016–2021	5	–	NA	Russian Banks	Russia	Logit Model	NA
Alzugaiby et al. (2021)	2021	1985–2016	31	–	8,260 small banks, 12,977 medium banks, 7,210 large banks	Small, Medium, and Large Banks	USA	Multivariate Regression Model	61
Durand et al. (2021)	2021	2000–2018	18	Basel II (2007), Basel III (2013), IFRS 9 (2018)	4,707 banks (USA), 3,529 banks (Europe)	European and US Banks	USA and 12 European countries: Austria, Belgium, Cayman Islands, Denmark, France, Germany, Ireland, Italy, Luxembourg, Netherlands, Spain, United Kingdom	Logit Model; Random Forest; Artificial Neural Networks	24
Obeid (2021)	2021	2005–2015	10	Basel II (2007), Basel III (2013)	40	Commercial Banks	Arabia	Logit Model	22
Bräuning et al. (2020)	2020	2014–2016	2	Basel III (2013)	3,000	European LSIs	Europe	Machine Learning	3,000
Gutiérrez-López and Abad-González (2020)	2020	2014–2018	4	Basel III (2013), IFRS 9 (2018)	45	European Union (EU) Banks	Europe	Multilevel Modeling	17
Shrivastava et al. (2020)	2020	2000–2017	17	–	56	Private and Public Sector Banks	India	Logit Model; Random Forest; AdaBoost	27
Casabianca et al. (2019)	2019	1970–2017	47	–	–	Not Disclosed	100 countries (33 advanced, 67 emerging)	Machine Learning and Logit Models	13
Cheong and Ramasamy (2019)	2019	2004–2010	6	–	536	Not Disclosed	USA	Logit Model	18
Ferriani et al. (2019)	2019	2008–2016	8	Bcc (2016), GACS/NPL (2016), Basel III (2013)	355	Less Significant Institutions (Bcc)	Italy	Logit Model	118
Shibitov and Mamedli (2019)	2019	2014–2018	4	–	NA	Russian Banks	Russia	Random Forest; Gradient Boosting; Neural Networks; Logit Model	69
Baker (2018)	2018	2008–2017	9	–	14	Deposit-Taking Institutions	Jamaica	Logit Model	13

(*continued on next page*)

Table A.7 (*continued*).

Paper	Year	Sample Window	Span (years, inclusive)	Reforms and Measures	Number of Banks in the Sample	Types of Banks	Geographic Area of the Sample	Methodology	Number of Variables Analyzed
Le and Viviani (2018)	2018	2010–2016	5	–	3,000	Commercial Banks	USA	Traditional Statistical Techniques (Discriminant Analysis, Logit); Machine Learning (ANN, SVM, k-NN)	31
Pappas et al. (2017)	2017	1995–2010	15	–	421	Islamic Banks and Conventional Banks	Middle and Far Eastern countries	Cox Proportional Hazards Model	33
Pruitt (2017)	2017	2012–2015	3	–	250	Commercial Banks	USA	Logit Model	3
Masood et al. (2016)	2016	2015	1	–	6	Islamic Banks	Pakistan	CAMELS Rating Model	8
Wang and Jiang (2016)	2016	2003–2013	10	–	775	Public Banks	Developing and Developed Countries	Logit Model	11
Calabrese and Giudici (2015)	2015	1996–2011	15	Basel II (2007)	783	Not Disclosed	Italy	Generalized Extreme Value (GEV) Model; Logit Model	14
Chiaramonte et al. (2015)	2015	2001–2010	9	Basel II (2007)	3,242	Not Disclosed	12 European Countries	Z-Score; Probit Model; Tobit Model	NA
Kenneth and Adeniyi (2014)	2014	2006–2010	5	–	9	Not Disclosed	Nigeria	Multiple Discriminant Model	6
Betz et al. (2014)	2014	2000Q1–2013Q2	13	Basel II (2007), Basel III (2013)	546	Not Disclosed	All EU countries except: Cyprus, Estonia, Lithuania, Romania	Logit Model	27
Moody's Analytics (2016)	2012	1988–2012	24	Basel I (1988), Basel II (2007)	1,205+	Not Disclosed	90+ Countries	Probit Model	NA
Montgomery et al. (2005)	2005	Indonesia (1997–2003); Japan (1978–2001)	Indonesia (6); Japan (23)	–	Indonesia (148); Japan (156)	Indonesia: Commercial Banks, State-Owned Banks, Private National Banks, Foreign Exchange and Non-Exchange Banks, Regional Development Banks, Joint-Venture Banks, Foreign Banks. Japan: Commercial Banks, City Banks, Long-Term Credit Banks, Trust Banks, Regional I and II Banks.	Japan and Indonesia	Logit Model	18

[a] GACS = Guarantees on the Securitization of Bad Loans (state guarantees to facilitate NPL disposals in Italy).[b] IFRS 9 = International Financial Reporting Standard, effective Jan. 1, 2018; introduces the Expected Credit Loss (ECL) model.[c] Cura Italia (Mar. 2020) = COVID-era measures supporting the health system and the economy.[d] LSIs = Less Significant Institutions. G-SII = Global Systemically Important Institutions. NA = not available/applicable.[e] *Recency key:* Rows with period end in 2020–2025 reflect post-IFRS 9 and post-COVID conditions; earlier rows are retained as seminal or methodological baselines.[f] Span (years, inclusive) counts unique calendar years in the sample window (end - start + 1). Quarterly windows (e.g., 2000Q1-2013Q2) are counted by calendar years covered. Multicountry rows report spans per panel.

Table A.8
Overview of closest studies on PD for Italian banks (post-IFRS 9/Basel III setting; explicit treatment of state aid and BCC/GACS measures).

Key Features	Calabrese and Giudici (2015)	Ferriani et al. (2019)	Chironna and Orlando (2023)	Predicting Bank Defaults in Italy (present paper)
Sample Window	1996–2011	2008–2016	2010–2020	2010–2020
Span (years, inclusive)	16	9	11	11
State aid in the period	State aid for very limited amounts (relative to asset size)	State interventions are not considered distress events because no Italian LSIs were affected in the sample period.	(1) 25 Jun 2017: €4.785bn cash + €12bn state guarantees for market exit of Banca Popolare di Vicenza and Veneto Banca (to Intesa Sanpaolo). (2) 4 Jul 2017: €5.4bn Italian aid to MPS approved by the European Commission.	(1) 25 Jun 2017: €4.785bn cash + €12bn state guarantees for market exit of Banca Popolare di Vicenza and Veneto Banca (to Intesa Sanpaolo). (2) 4 Jul 2017: €5.4bn Italian aid to MPS approved by the European Commission.
Data frequency	Annual	Quarterly	Annual	Annual
Outlier detection and treatment	Not disclosed	Not disclosed	Outliers removed (see Table 2).	Outliers removed (see Table 2).
Default/Distress definition	Two definitions of failure: (i) actual defaults (bankruptcy, dissolution, liquidation); (ii) defaults plus M&A events.	Extended definition including: compulsory administrative liquidation; extraordinary/temporary administration; voluntary liquidation; merger in distress; disposal of assets; resolution; interventions of FITD and FGDCC (also preventive); notification of financial deterioration to the ECB.	One of: merger/acquisition; bail-out (state aid); bail-in; intervention by FITD; extraordinary administration (ECB/BoI); compulsory administrative liquidation (winding up).	One of: merger/acquisition; bail-out (state aid); bail-in; intervention by FITD; extraordinary administration (ECB/BoI); compulsory administrative liquidation (winding up).
State aids included in distress definition	Not included	Not included	Included	Included
Reforms and measures	Basel II (Jan. 2007)	BCC reform (2016); GACS/NPL (2016); Basel III (Dec. 2013)	BCC reform (2016); GACS/NPL (2016); Basel III (Dec. 2013); IFRS 9 (Jan. 2018); *Cura Italia* (Mar. 2020)	BCC reform (2016); GACS/NPL (2016); Basel III (Dec. 2013); IFRS 9 (Jan. 2018); *Cura Italia* (Mar. 2020)
Number of banks in the sample	783	355	100	100
Types of banks	Large number; unlisted and small banks	LSIs (BCC)	LSIs (BCC); Cooperative (Popolari); SpA; foreign branches; listed; G-SII	LSIs (BCC); Cooperative (Popolari); SpA; foreign branches; listed; G-SII
Geographic area	Italy	Italy	Italy	Italy
Methodology	Generalized Extreme Value (GEV); Logit Model	Logit Model	Conventional (Logit Model); Machine Learning (GLM, CART, Random Forest)	Logit Model; Probit Model; Linear Probability Model; Panel Logit; Machine Learning (CART, Random Forest); Logit with Diversification Indices; Logit with Interaction Term; Generalized Additive Model (GAM)
Number of variables analyzed	14	118	72+	72+ micro; 3 macro (+ lags); 4 Diversification Indices; 1 Interaction Term
Types of variables	Macro; Micro	Sectoral; Macro; Micro	Micro (financial statements)	Micro (financial statements); Macro (CPI, GDP growth, interest rates); Diversification Indices (Earnings Assets; Income; Deposits; Loan Portfolio); Interaction terms

(*continued on next page*)

Table A.8 (*continued*).

Key Features	Calabrese and Giudici (2015)	Ferriani et al. (2019)	Chironna and Orlando (2023)	Predicting Bank Defaults in Italy (present paper)
Main explanatory variables	Intercept; Loan-loss reserves/Loans; Loans/Total assets; Equity/Total assets; Tier 1 ratio; Total capital ratio; Liquid assets/Deposits; ROA; GDP; inflation; unemployment; interest rate	Model 1 (4Q/6Q): CET1; NPL; interconnectedness; loan value adjustments/TA; Total assets; ROA; net income/operating costs; customer deposits/ administrative expenses; gross operating profit/ managed resources; demand assets/demand liabilities; loans to customers/total financing; highly liquid assets/total deposits; funding gap; board turnover; number of directorships; BCC dummy; constant	Pr05 (best): constant; Equity/Assets; ROA; BCC (dummy). ROSE GLM: constant; ROA; net impairments (losses) on a)+b)/credits vs customers	Pr05 (best): constant; Equity/Assets; ROA; BCC (dummy). ROSE GLM: constant; ROA; net impairments (losses) on a)+b) / credits vs customers. Logit with Diversification Indices: constant; Earnings Assets; Income; Deposits; Loan Portfolio. Logit with Interaction: constant; ROA; lagged (two-period) GDP growth; Equity/Assets; BCC (dummy); interaction ROA ×lagged two-period GDP growth. GAM: constant; ROA; BCC (dummy); smooth terms for ROA and lagged two-period GDP growth
Variable selection	VIF	Coverage of risk areas and statistical robustness; weak/unstable regressors excluded from final specifications	Conventional (see Table 4); ML (adaptive lasso)	Conventional (see Table 4); ML (adaptive lasso); Johansen cointegration analysis; Granger causality test; interaction and nonlinear-effects analysis
Number of terms	Without M&A: 12 With M&A: 12	M-1-4Q: 17 M-1-6Q: 15	Pr05: 4 ROSE GLM: 3	Pr05: 4 ROSE GLM: 3 Logit with Diversification Indices: 5 Logit with Interaction: 6 GAM: 4
AUC	GEV (F): 0.7134; GEV (F+M): 0.8846; Logit (F): 0.7889; Logit (F+M): 0.9380	Model 1 (4Q): 0.9148 Model 1 (6Q): 0.8905	Pr05: 0.856 ROSE GLM: 0.846	Pr05: 0.856 ROSE GLM: 0.846
Forecasting horizons	1-year ahead	4Q and 6Q ahead	1-year ahead	1-year ahead
Limitations	No state aids in distress definition; outlier handling not disclosed; limited set of 14 variables	No state aids included; LSIs only; outlier handling not disclosed; 4–6Q horizon may be short for early supervisory actions; methodology restricted to Logit Model; limited number of observation years	No macro variables; limited observation years; limited number of banks in sample	Limited observation years; limited number of banks in sample

[a] GACS and IFRS 9 as defined in Table A.7 notes. LSIs = Less Significant Institutions; G-SII = Global Systemically Important Institutions.
[b] *Span (years, inclusive)* counts unique calendar years in the sample window (end − start + 1).
[c] *Recency key:* Studies covering 2018–2020 reflect the IFRS 9 environment; earlier studies serve as methodological baselines.

Appendix C. Handling data uncertainty and robustness

Real-world financial data are inevitably subject to multiple sources of uncertainty, including measurement errors, missing values, and stochastic variability. To address these issues, our preprocessing pipeline incorporates systematic cleaning, winsorization, and explicit handling of missing observations, thereby reducing the impact of extreme or unreliable entries.

To mitigate measurement errors, our data entry process employed a double-check structure in Microsoft Excel to identify discrepancies in reported financial statement figures. When an error was flagged, the value was verified against the prior-year figure from the most recent financial statement. We then consulted the accompanying notes to determine the reason and timing of any restatement. If a restatement was confirmed, the updated value was imputed to ensure the use of the most accurate and current data.

Missing values were handled as follows. In a dataset consisting of 1059 financial statements with 72 performance indicators (for a total of 76,248 value fields), only 54 missing values were detected, representing 0.071% of the dataset. These were imputed using the median rather than the mean. As argued by Matyja and Siminski (2014), mean substitution may yield non-existent or non-interpretable values, whereas median imputation avoids this drawback. Their results also show that, for small proportions of missing data, median imputation provides superior performance.

Table B.9
Geographical Area: Sample vs Italian Banking System.
Source: Bank of Italy.

Geographical area	Sample							N.(Italy)							Sample/N.(Italy)		
	NBA'10	W'10	NBA'20	W'20	ND'10-'20	WD'10-'20	WD'10	B'10	WB'10	B'20	WB'20	D'20-'10	WD'20-'10	WD'10	WS'10	WS'20	WSD'10-'20
North-west	24	.240	20	.253	−4	.190	−0.167	214	.282	167	.352	−47	.164	−0.220	.112	.120	.085
North-east	28	.280	23	.291	−5	.238	−0.179	241	.317	127	.268	−114	.399	−0.473	.116	.181	.044
Center	18	.180	13	.165	−5	.238	−0.278	160	.211	89	.188	−71	.248	−0.444	.113	.146	.070
South	23	.230	19	.241	−4	.190	−0.174	105	.138	69	.146	−36	.126	−0.343	.219	.275	.111
Islands	7	.070	4	.051	−3	.143	−0.429	40	.053	22	.046	−18	.063	−0.450	.175	.182	.167
Total	100	1.000	79	1.000	−21	1.000	−0.210	760	1.000	474	1.000	−286	1.000	−0.376	.132	.167	.073

Note. Names and meanings of columns:
Sample
NBA '10: Number of Banks Analyzed (2010); W '10: Number of Banks Analyzed /Sample (2010); NBA '20: Number of Banks Analyzed (2020); W '20: Number of Banks Analyzed /Sample (2020); ND '10-'20: Number of Default for the area (2010–2020); WD '10-'20: Default For Area Of the Sample/Total Def. Of The Sample (2010–2020); WD '10: Number of Default for the area (2010–2020)/Number of Banks Analyzed (2010);
N.(Italy)
B'10: Banks (2010); WB'10: B'10/Total B'10; B'20: Banks (2020); WB'20: B'20/Total B'20; D'20-'10: Number of Banks Disappeared 2020–2010; WD'20-'10: D'20-'10 for area/Number of Banks Disappeared 2020–2010; WD'10: D'20-'10 for area/Banks 2010;
Sample/N.(Italy)
WS'10: NBA '10/B'10; WS'20: NBA '20/B'20; WSD'10-'20: ND'10-'20/D'20-'10

Table B.10
Legal Form: Sample vs Italian Banking System.
Source: Bank of Italy.

Legal Form	2010				2020				Diff. 2010–2020	
	B'10	WB'10	BS'10	WS'10	B'20	WB'20	BS'20	WS'20	DIF.B'20-B'10	DIF./B'10
Spa Banks	233	.307	25	.107	123	.259	14	.114	−110	−0.472
Cooperative Banks (Banche Popolari)	37	.049	10	.270	21	.044	7	.333	−16	−0.432
Mutual Banks (BCCs)	415	.546	63	.152	248	.523	56	.226	−167	−0.402
Foreign Bank Branches	75	.099	2	.027	82	.173	2	.024	7	0.093
Total	760	1.000	100	.132	474	1.000	79	.167	−286	−0.376

Note. In our sample, we consider Credit Agricole Italia and Deutsche Bank S.p.a. as foreign bank branches. Names and meanings of columns:
2010
B'10: Number of Banks in Italy 2010; WB'10: B'10/Total Banks in Italy 2010; BS'10: Number of Banks in the Sample 2010; WS'10: BS'10/B'10;
2020
B'20: Number of Banks in Italy 2020; WB'20: B'20/Total Banks in Italy 2020; BS'20: Number of Banks in the Sample 2020; WS'20: BS'20/B'20;
Diff. 2010–2020
DIF.B'20-B'10: Difference B'20-B'10; DIF./B'10: DIF.B'20-B'10/B'10.

In addition, both conventional econometric models and machine-learning classifiers are validated using cross-validation and resampling procedures to account for noise and sampling variability. We further calibrate probabilistic outputs (e.g., predicted default probabilities) to enhance interpretability and reliability. Finally, we conduct extensive robustness checks — including alternative model specifications, sub-sample analyses, and placebo tests — to ensure that the results are not driven by spurious patterns or model-specific artefacts. Collectively, these steps improve the resilience of our framework to data uncertainty and reinforce the credibility of the empirical findings.

Appendix D. Regression tables and ROC

We present the results of multiple regression models applied to two datasets — DS(B100) and DS(B73) — using different strategies for handling class imbalance: imbalanced, over-sampling, under-sampling (Both), and ROSE (Random Over-Sampling Examples).

Table D.12 compares the performance of Generalized Linear Models (GLM) trained with the Both and ROSE sampling techniques, reporting coefficient estimates, standard errors, z-values, and p-values. Notably, ROSE outperforms Both with a higher AUC (0.846) and Youden's J (0.692), indicating improved classification capability.

Table D.13 summarizes the GLM performance metrics on the DS(B100) dataset across all four balancing methods. ROSE yields the highest AUC and Youden's J, confirming its effectiveness. Fig. D.13 visualizes the ROC curves for the GLM models on DS(B100), illustrating superior performance for the ROSE and Both methods.

Table D.14 shows the GLM results for the DS(B73) dataset. Again, the Both and ROSE methods perform well, with Both achieving near-perfect metrics. Fig. D.14 provides the ROC curves for GLM models on DS(B73), further supporting the performance of Both and ROSE.

Tables D.15 and D.16 present CART model results for DS(B100) and DS(B73), respectively. In both datasets, the Both method consistently achieves the highest Youden's J and AUC, particularly on DS(B100) where AUC reaches 0.996. Figs. D.15 and D.16 show corresponding ROC curves for CART models. The dominance of the Both method is visually apparent.

Tables D.17 and D.18 report Random Forest results for DS(B100) and DS(B73). The Both method achieves perfect scores (AUC and Youden's J of 1.000) on DS(B100), while ROSE performs strongly on DS(B73).

Table B.11
Descriptive Statistics (after filtering for outliers)

Variable	Variable description	Def	Obs	Mean	Std. Dev.	Min	Max
Def	Default Flag**	1	946	0.02	0.15	0	1
BCC	Dummy for small bank (Banca di Credito Cooperativo, BCC); see EACB (2016)	−0.1107*	1045	0.48	0.50	0	1
I1_ROE	Roe = Profit/Net Equity	−0.3260*	1045	0.01	0.1	−0.89	0.28
I2_PN/TA	Net Equity/Total Assets = Equity/Total Assets = Book ValueTotal Assets	−0.1068*	1045	0.09	0.04	0	0.43
I3_PN/CVC	Net Equity/Loans Vs Customers = Equity/Total Loans	−0.0367	1045	0.18	0.34	0	10.09
I4_ROA	ROA = Profit/Total Assets	−0.3050*	1045	0	0.01	−0.06	0.02
I4_ROAd	Change in ROA: ROA(t) - ROA(t-1)	−0.0779*	948	0	0.01	−0.03	0.04
I5_LA/TA	Liquidity = Liqudity Assets/Total Assets	0.0071	1045	0.01	0.01	0	0.16
I6_P/TA	Loans = Loans/Total Assets	0.0671*	1045	0.62	0.16	0	0.95
I7_D/TA	Deposits Vs Customers/Total Assets	0.0175	1045	0.55	0.14	0	0.96
I8_MI/MID	Interest Margin/Brokerage Margin	−0.0725*	1045	0.6	0.14	0	1.17
I9_RAPSI/TA	Result Before Extraordinary Income and Taxes/Total Assets	−0.2862*	1045	0	0.01	−0.07	0.04
I10_T1CR	Tier 1 capital ratio (Tier 1 capital _ Risk weighted assets)	−0.0841*	1045	0.17	0.08	0	0.64
I11_TCR	Total capital ratio (Regulatory capital including TIER 3 _ Risk weighted assets)	−0.0761*	1045	0.18	0.07	0	0.69
I12_RRV_C/PN	Net Value Adjustments _ Write-backs Due to Impairment of: "Loans"/Net Equity	−0.2602*	1045	−0.08	0.1	−1.31	0.1
I13_RLG	Rate of Loan Growth	−0.0115	1045	0.2	2.73	−1	85.95
I14_RRV_C/CVC	Net adjustments _ write-backs for deterioration of Loans/Loans	−0.0258	1045	0	0	−0.03	0.03
I15_CCN/TA	CCN (Interest-bearing assets - Interest-bearing liabilities) (Net working capital NWC)/Total Assets	−0.1393*	1045	0.07	0.04	−0.48	0.44
I16_CCN/PN	Net working capital/Net Equity	−0.1543*	1045	0.8	0.29	−4.45	2.14
I17_CVC/DVC	Receivables from customers/Deposits from customers	0.0422	1045	1.17	0.48	0	6.99
I18_RRV_C/CCN	Net adjustments _ write-backs for impairment of: "Loans"/Net working capital	0.0073	1045	0.01	0.04	−0.57	0.56
I19_CVC/AF	Receivables from customers/interest-bearing assets	0.0827*	1045	0.65	0.17	0	0.97
I20_TDD/AF	Debt Securities/Interest-bearing Assets	−0.0957*	1045	0.24	0.14	0	0.84
I21_P/AF	Loans/Interest-bearing activities	0.0042	1045	0	0.01	0	0.14
I22_DVB/PO	Payables to Banks/Expensive Liabilities	−0.0036	1045	0.15	0.11	0	0.77
I23_DVC/PO	Payables to customers/Onerous liabilities	0.0053	1045	0.63	0.17	0	1
I24_DT/PO	Payables for Securities/Onerous Liabilities	0.0012	1045	0.18	0.12	0	0.56
I25_RRV_130/TA	Net adjustments _ write-backs for impairment of loans (V. 130 EC)/Total Assets	−0.2181*	1045	−0.01	0.01	−0.05	0
I26_RRV_AM/TA	Net adjustments _ write-backs on tangible assets (V.200 EC)/Total Assets	−0.0322	1045	0	0	−0.01	0
I27_RRV_AI/TA	Net adjustments _ write-backs on intangible assets (V.210 CE)/Total Assets	−0.052	1045	0	0	−0.01	0.01
I28_RVA/TA	Value adjustments of goodwill/Total Assets	−0.0776*	1045	0	0	−0.02	0
I29_TRRV/TA	Total adjustments _ write-backs/Total assets	−0.2209*	1045	−0.01	0.01	−0.06	0
I30_TRRV/PN	Total adjustments _ write-backs/Net equity	−0.2585*	1045	−0.11	0.12	−1.5	0.06
I31_TA/PN	Total Assets/Net Equity	0.1149*	1045	12.9	7	0	89.84
I32_TAT/PN	Total Tangible Assets/Tangible Net Equity	0.1036*	1045	13.69	7.59	0	93.03
I33_MI/TA	Interest Margin/Total Assets	−0.0745*	1045	0.02	0.01	0	0.06
I34_MID/TA	Brokerage Margin/Total Assets	−0.0535	1045	0.03	0.01	0	0.1
I35_RG/TA	Operating Profit/Total Assets	−0.1188*	1045	0.01	0.01	−0.03	0.04
I36_RAPSI/PNT	Result Before Extraordinary Income and Taxes/Tangible Net Equity	−0.2418*	1045	0.02	0.12	−1.44	0.43
I37_RAI/TA	Pre-Tax Result/Total Assets	−0.2744*	1045	0	0.01	−0.07	0.04
I38_RN/RIS(140)	Net Result/Reserves (140)	0.0404	1045	−0.44	17.62	−565.37	53.62
I39_RG/PN	Operating Profit/Net Equity	−0.0561	1045	0.12	0.07	−0.27	0.58
I40_RAPSI/PN	Result Before Extraordinary Income and Taxes/Net Equity	−0.2969*	1045	0.02	0.12	−1.33	0.43
I41_RAI/PN	Pre-Tax Result/Net Equity	−0.2859*	1045	0.02	0.12	−1.28	0.43
I42_ROTE	ROTE = Net Result/Tangible Net Equity	−0.3088*	1045	0.01	0.11	−0.96	0.3
I43_RG/MID	Operating Profit/Brokerage Margin	−0.1432*	1045	0.32	0.17	−1.35	1.5
I44_RAPSI/RG	Result Before Extraordinary Income/Operating Result	−0.0898*	1045	0.31	2.89	−27.19	75.81
I45_RAI/RAPSI	Result Before Taxes/Result Before Extraordinary Income and Taxes	−0.0073	1045	1.11	2.13	−2.69	68.22
I46_RN/RAI	Net Result/Pre-Tax Result	−0.0132	1045	1.43	22.77	−66.67	731.98
I47_MI/SA	Interest Margin/Administrative Expenses	0.1090*	1045	−0.88	0.26	−2.72	0
I48_MID/SA	Brokerage/Administrative Expenses	0.0897*	1045	−1.45	0.33	−3.64	0
I49_MI/CO	Interest Margin/Operating Costs	0.1161*	1045	−0.94	0.3	−2.72	1.23
I50_MID/CO	Brokerage Margin/Operating Costs	0.0991*	1045	−1.55	0.39	−3.67	1.99
I51_BKS_LN(TA)	Bank Size (Proxy Variable) = Ln (TA)	0.06	1044	14.8	2.66	8.85	24.65
I52_RN/MID	Net Result/Brokerage Margin	−0.3153*	1045	0.05	0.26	−3.22	0.74
I53_RN/CO	Net Income/Operating Costs (Net Income _ Operating Costs)	0.2546*	1045	−0.12	0.35	−2.45	2.67
I54_CO/MID	Operating Costs/Brokerage Margin	−0.1333*	1045	−0.67	0.19	−2.35	0.63
I55_RRVsC/MID	Net adjustments _ write-backs for impairment of: a) Loans/Brokerage margin	−0.2492*	1045	−0.22	0.21	−1.91	0.19
I56_MID/SP	Brokerage margin/Personnel expenses	0.057	1045	−2.47	1.16	−8.44	4.92
I57_Fc/Ac	Fc/Ac = (PN + DvB + DvC)/(DL + CvB + CvC)	−0.0651*	1045	1.14	0.37	0	4.14
I58_RAV/AI	RAV/AI = Value adjustments of goodwill/intangible assets	−0.0102	1045	−0.69	14.07	−440.76	0
I59_AM/TA	Tangible Assets/TA	0.043	1045	0.01	0.01	0	0.06
I60_AF/(TA-PN)	Financial assets/(TA-PN)	−0.1023*	1045	0.25	0.16	0	0.9
I61_DVC/(TA-PN)	Customer deposits/(TA-PN)	0.0002	1045	0.61	0.16	0	0.98
I62_DVC/SA	Deposits Vs Customers/Administrative Expenses	−0.0014	1045	−27.73	10.83	−149.74	0
I63_DVB/(PN+DVB+DVC)	Payables Vs Banks/(Net Equity+Deb.Vs.Banks+Deb.Vs. Customers)	0.0067	1045	0.18	0.12	0	0.88
I64_P/(PN+DVB+DVC)	Loans/(PN + Debt to Banks + Debt to Clients)	0.0592	1045	0.82	0.26	0	1.8
I65_CDL/DVC	Cash and cash equivalents/payables to customers	0.0051	1045	0.02	0.02	0	0.25
I66_(CVC-RRV130)/DVC	(Receivables to Customers - Adj. Rep. To Net Values Per Det. (V.130))/Payables Vs Customers	0.0405	1045	1.2	0.47	0	7.03
I67_CVB/AF	Loans Vs Banks/Interest-bearing Assets	0.0016	1045	0.09	0.1	0	0.79
I68_RRV130/CVC	Net impairment (losses)/reversals on (a)+(b) (V.130))Credits Vs Customers	−0.0836*	1045	−0.01	0.02	−0.5	0.02
I69_RN/(TA-PN)	Net Result/(TA-PN) or Net Result/Total Liabilities	−0.2973*	1045	0	0.01	−0.06	0.03
I70_IP/(TA-PN)	Interest expense/(TA-PN) or Interest expense/Total Liabilities	0.0173	1045	−0.01	0.01	−0.04	0
I71_RRV130/PNT	Adjustments Rep. A Val. Net for Det. (V.130))/Tangible Net Equity	−0.2455*	1045	−0.09	0.11	−1.45	0.1
I72_RRV130/MID	Adjustments Rep. A Val. Net for Det. (V.130))/Brokerage margin	−0.2497*	1045	−0.23	0.21	−1.96	0.17

[a] * Significant at 5%.
[b] ** The number of observations for 'Def' is less by almost 100 observations (i.e. 1045-946) due to one year shift when predicting PD for 100 banks one year ahead.

Fig. D.17 and Fig. D.18 display the ROC curves for the Random Forest models on DS(B100) and DS(B73), respectively, with Fig. D.17 highlighting the superior performance of the Both method.

In summary, across models and datasets, the Both and ROSE techniques generally lead to significantly better predictive performance, as evidenced by improvements in AUC, sensitivity, and Youden's J.

Table D.12
Machine Learning Results: Both vs ROSE.

	Model							
Variable	Both				ROSE			
	Estimate	Std.Error	z Value	Pr(>abs(z))	Estimate	Std.Error	z Value	Pr(>abs(z))
Constant Intercept	−2.3034	1.776	−12.972	$<2e-16^{***}$	−1.0987	0.1273	−8.634	$<2e-16^{***}$
I4 ROA ROA = Profit/Total Assets	−141.5124	19.0504	−7.428	$<1.1e-13^{***}$	−60.4569	6.7506	−8.956	$<2e-16^{***}$
I68 RRV130/CVC Net imp. (losses) reversals on a)+b)/Credits Vs Customers	−25.4647	9.2292	−2.759	$<0.00508^{**}$	−25.5172	4.2997	−5.935	$<2.94e-09^{***}$
AUC	0.779				0.846			
Youden's J	0.558				0.692			

[a] Significance: 0. '***'; 0.001 '**'; 0.01 '*'
z Value is the ratio of the estimated coefficient to its standard error. A higher absolute value of z value indicates that the estimated coefficient is more statistically significant. Pr(> abs(z)) represents the *p*-value for each coefficient, which is the probability of observing a z value as extreme. A lower *p*-value indicates that the coefficient is more statistically significant.
The larger dataset of 100 banks favors the GLM models.
The best performance, with the machine learning approach, is achieved by the GLM ROSE algorithm, which yields a Youden's J value of 0.692 and an AUC of 0.846. The next best model is also the GLM regression using the original Both data approach with a Youden's J value of 0.558 and an AUC of 0.779.

Table D.13
GLM Regression DS(B100).

Solution to unbalanced class	Accuracy	Sensitivity (EII°T)	Specificity	P-Value	Kappa	AUC	Youden's J
Imbalanced	0.970	1.000	0.000	0.599	0.000	0.500	0.000
Over Sampling	0.970	0.987	0.429	0.599	0.446	0.708	0.415
Both (Over-Under Sampling)	0.974	0.987	0.571	0.447	0.558	0.779	0.558
ROSE	0.970	0.978	0.714	0.599	0.573	0.846	0.692
Mean	0.971	0.988	0.429	0.561	0.394	0.708	0.416
Median	0.970	0.987	0.500	0.599	0.502	0.744	0.487

[a] GLM regression model results with the four DS(B100) dataset balancing solutions.

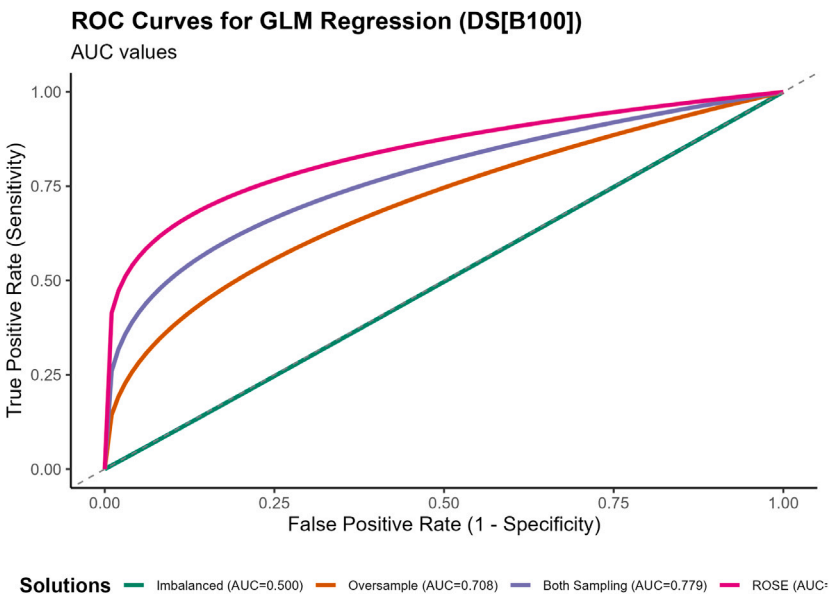


Fig. D.13. ROC Curve for the GLM regression model results with the four DS(B100) dataset balancing solutions.

Table D.14
GLM Regression DS(B73).

Solution to unbalanced class	Accuracy	Sensitivity (EII°T)	Specificity	P-Value	Kappa	AUC	Youden's J
Imbalanced	0.981	1.000	0.000	0.647	0.000	0.500	0.000
Over Sampling	0.975	0.975	1.000	0.817	0.589	0.987	0.975
Both (Over-Under Sampling)	0.994	0.994	1.000	0.196	0.854	0.997	0.994
ROSE	0.975	0.975	1.000	0.817	0.529	0.987	0.975
Mean	0.981	0.986	0.750	0.619	0.493	0.868	0.736
Median	0.978	0.984	1.000	0.732	0.559	0.987	0.975

[a] GLM regression model results with the four DS(B73) dataset balancing solutions.

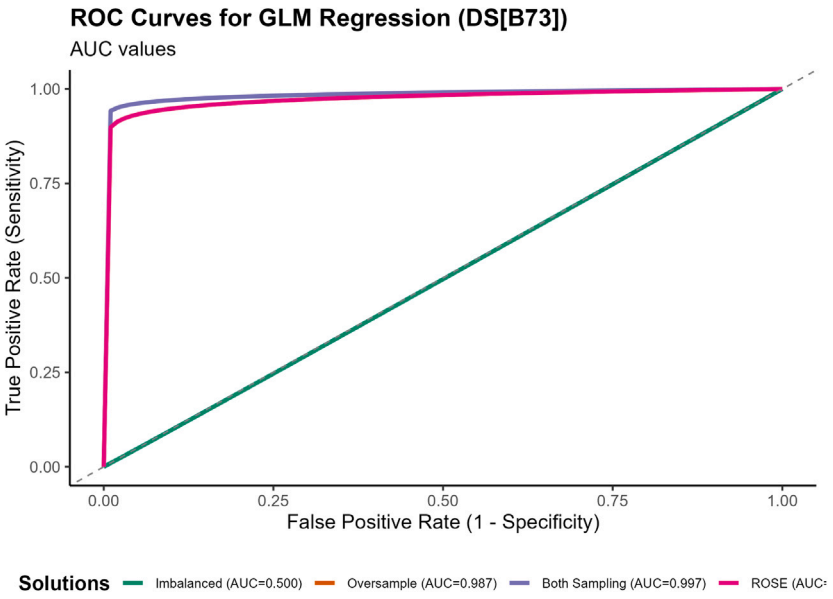


Fig. D.14. ROC Curve for the GLM regression model results with the four DS(B73) dataset balancing solutions.

Table D.15
CART model results for DS(B100).

Solution to unbalanced class	Accuracy	Sensitivity (EII°T)	Specificity	P-Value	Kappa	AUC	Youden's J
Imbalanced	0.970	0.991	0.286	0.559	0.349	0.641	0.277
Over Sampling	0.957	0.978	0.286	0.905	0.263	0.632	0.263
Both (Over-Under Sampling)	0.991	0.991	1.000	0.028	0.871	0.996	0.991
ROSE	0.917	0.924	0.714	1.000	0.313	0.819	0.638
Mean	0.959	0.971	0.571	0.633	0.449	0.772	0.542
Median	0.963	0.984	0.500	0.752	0.331	0.730	0.457

[a] CART model results with the four balancing solutions on DS(B100) datasets.

Table D.16
CART results for DS(B73).

Solution to unbalanced class	Accuracy	Sensitivity (EII°T)	Specificity	P-Value	Kappa	AUC	Youden's J
Imbalanced	0.981	1.000	0.000	0.647	0.000	0.500	0.000
Over Sampling	0.975	0.994	0.000	0.817	−0.009	0.503	−0.006
Both (Over-Under Sampling)	0.994	0.994	1.000	0.196	0.854	0.997	0.994
ROSE	0.926	0.930	0.667	0.999	0.266	0.799	0.597
Mean	0.969	0.979	0.417	0.665	0.278	0.700	0.396
Median	0.978	0.994	0.333	0.732	0.133	0.651	0.299

[a] CART model results with the four balancing solutions on DS(B73) datasets.

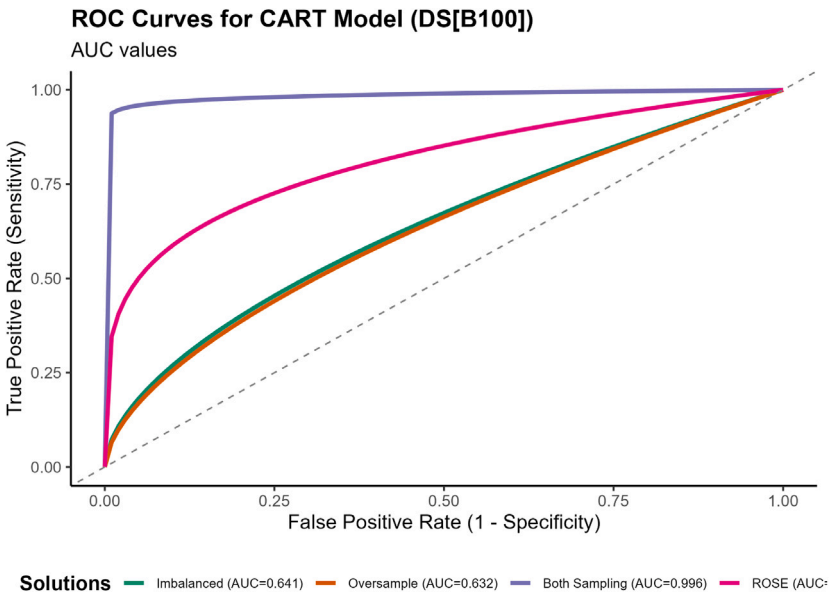


Fig. D.15. ROC Curve for the CART model results with the four DS(B100) dataset balancing solutions.

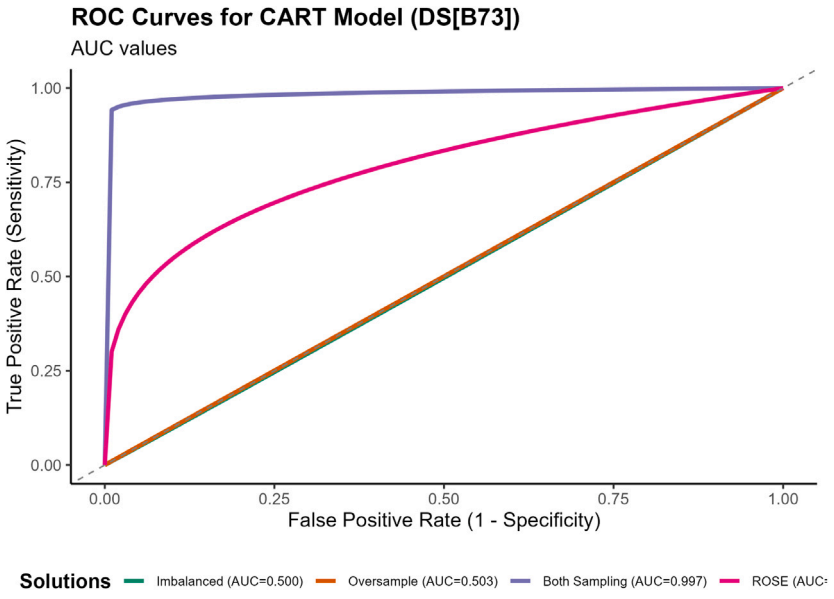


Fig. D.16. ROC Curve for the CART model results with the four DS(B73) dataset balancing solutions.

Table D.17
Random Forest DS(B100).

Solution to unbalanced class	Accuracy	Sensitivity (EII°T)	Specificity	P-Value	Kappa	AUC	Youden's J
Imbalanced	0.970	1.000	0.000	0.599	0.000	0.500	0.000
Over Sampling	0.965	0.991	0.143	0.731	0.185	0.567	0.134
Both (Over-Under Sampling)	1.000	1.000	1.000	0.001	1.000	1.000	1.000
ROSE	0.948	0.960	0.571	0.975	0.375	0.766	0.531
Mean	0.971	0.988	0.429	0.576	0.390	0.708	0.416
Median	0.967	0.996	0.357	0.665	0.280	0.667	0.332

[a] Random Forest results with the four balancing solutions on DS(B100) datasets.

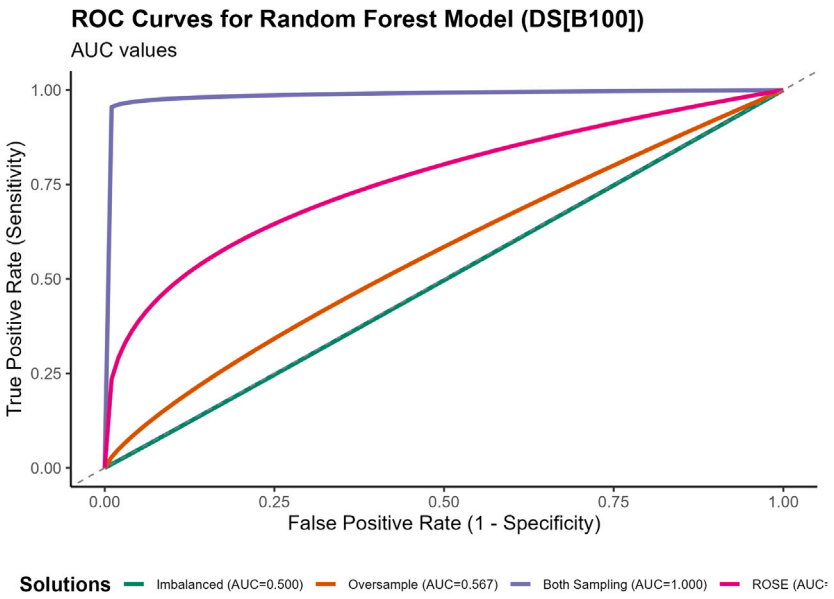


Fig. D.17. ROC Curve for the Random Forest model results with the four DS(B100) dataset balancing solutions.

Table D.18
Random Forests DS(B73).

Solution to unbalanced class	Accuracy	Sensitivity (EII°T)	Specificity	P-Value	Kappa	AUC	Youden's J
Imbalanced	0.981	1.000	0.000	0.647	0.000	0.500	0.000
Over Sampling	0.981	1.000	0.000	0.248	0.589	0.500	0.975
Both (Over-Under Sampling)	0.994	0.994	1.000	0.196	0.854	0.997	0.994
ROSE	0.963	0.962	1.000	0.968	0.486	0.981	0.962
Mean	0.980	0.989	0.500	0.518	0.335	0.745	0.489
Median	0.981	0.997	0.500	0.448	0.243	0.741	0.481

[a] Results of the random forests model with the four DS dataset balancing solutions (B73).

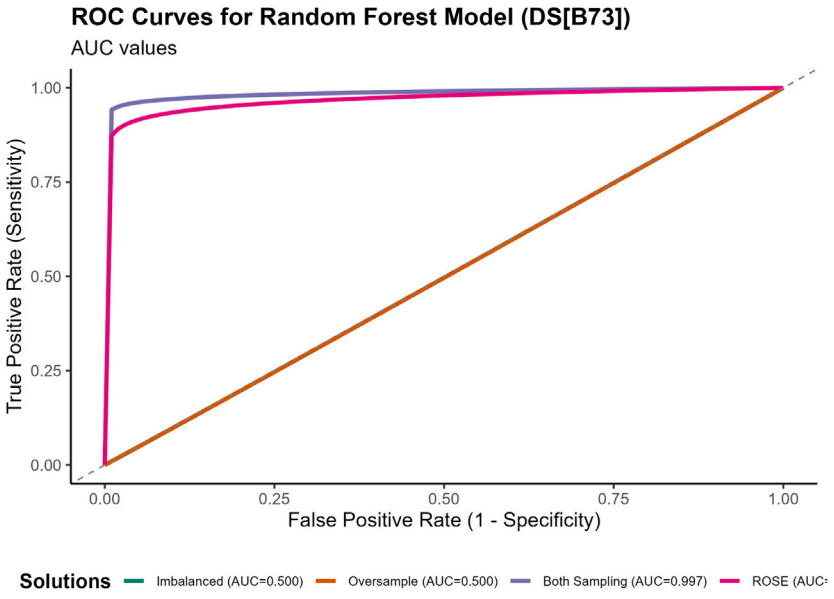


Fig. D.18. ROC Curve for the Random Forest model results with the four DS(B73) dataset balancing solutions.

Table E.19
Logistic Regression Results: Diversification Indices and Bank Default.

Variable	Coefficient Estimate	Significance Level
Intercept	−3.880	0.0132*
DIVassets	1.961	0.1949
DIVincome	1.721	0.2093
DIVdeposits	−2.089	0.4271
DIVloan	−2.200	0.1271

Note: None of the diversification indices are statistically significant. This suggests that they have limited explanatory power for predicting default events in a logistic regression framework. Although some coefficients are relatively large in magnitude (e.g., DIVloan), their lack of significance implies high variance and uncertainty in these effects.

Appendix E. Diversification indices and analysis

Following the methodology proposed by Manthoulis et al. (2021), we construct Herfindahl–Hirschman-type diversification indices for our sample banks. The general form is:

$$HHI = 1 - \sum_i x_i^2$$

where x_i denotes the share of the ith component in a given dimension (e.g., assets, income, deposits, loans). Based on data availability and financial structure, we define the following four indices:

Earning assets diversification (divassets).

$$DIVassets = 1 - \left(\left(\frac{\text{Loans}}{\text{Total Assets}} \right)^2 + \left(\frac{\text{Debt Securities}}{\text{Interest-bearing Assets}} \right)^2 + \left(\frac{\text{Financial Assets}}{\text{Total Assets} - \text{Equity}} \right)^2 \right)$$

Income diversification (divincome).

$$DIVincome = 1 - \left(\left(\frac{\text{Interest Margin}}{\text{Total Assets}} \right)^2 + \left(\frac{\text{Brokerage Margin}}{\text{Total Assets}} \right)^2 \right)$$

Deposit diversification (divdeposits).

$$DIVdeposits = 1 - \left(\left(\frac{\text{Deposits from Customers}}{\text{Total Liabilities}} \right)^2 + \left(1 - \frac{\text{Deposits from Customers}}{\text{Total Liabilities}} \right)^2 \right)$$

Loan portfolio diversification (divloan).

$$DIVloan = 1 - \left(\left(\frac{\text{Receivables from Customers}}{\text{Interest-bearing Assets}} \right)^2 + \left(\frac{\text{Loans to Banks}}{\text{Interest-bearing Assets}} \right)^2 \right)$$

These indices allow us to assess the role of diversification in influencing bank default probabilities, both in the short and long term.

E.1. Robustness checks: Diversification indices and default prediction

Before evaluating the effectiveness of the diversification indices, we apply two complementary empirical strategies. First, a logistic regression is used to assess whether the indices can serve as reliable predictors of bank default events in the short run. This regression focuses on whether diversification patterns, as measured at a given point in time, correlate with the probability of failure. Second, we investigate long-term equilibrium relationships by conducting a Johansen cointegration analysis. This method enables us to determine whether the diversification indices are structurally related to key financial stability indicators—namely, the ratio of interest-bearing assets to total assets and return on assets. By combining both approaches, we aim to distinguish between transitory effects and more persistent systemic linkages between diversification and financial fragility.

E.1.1. Predictive power of diversification: Logistic regression analysis

To evaluate the short-run predictive ability of the diversification indices, we estimate a logistic regression model where the probability of default is regressed on four diversification measures: DIVassets, DIVincome, DIVdeposits, and DIVloan. Table E.19 reports the estimated coefficients and significance levels from this logistic regression.

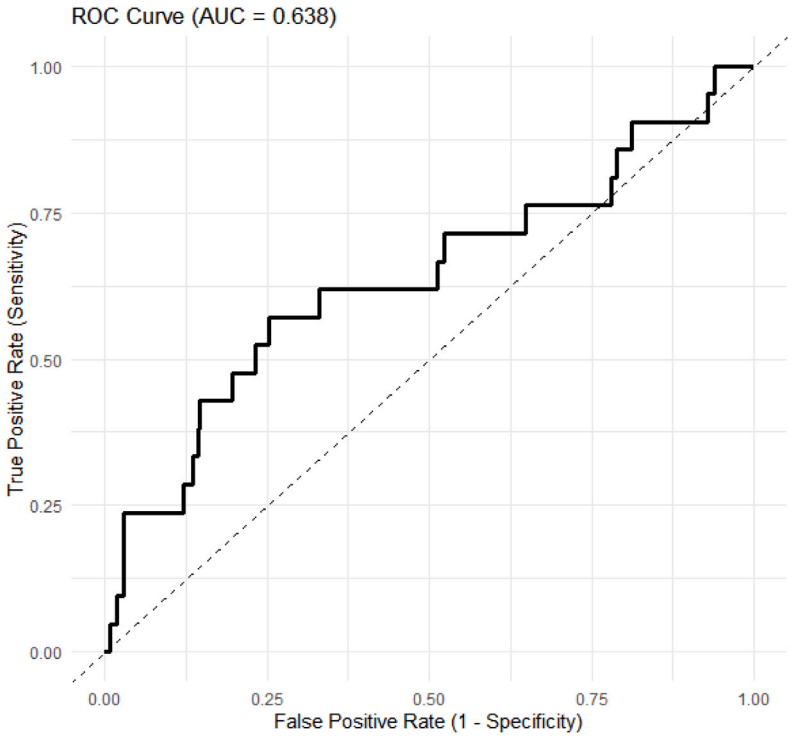


Fig. E.19. ROC Curve for the Logistic Regression Model Predicting Bank Default Based on Diversification Indices. AUC = 0.6379.

These results imply that, in isolation, diversification indices do not contribute meaningfully to short-term default prediction. This conclusion is further supported by the ROC curve presented in Fig. E.19, which visualizes the trade-off between sensitivity and specificity in classification.

As shown in the figure, the AUC equals 0.6379, indicating that the model performs only slightly better than random guessing in classifying defaulted banks.

The distributional properties of the diversification indices are presented in Figs. E.20 and E.21, as well as in Table E.20. These visualizations provide context for understanding their limited explanatory power.

Namely, Fig. E.20 shows that some indices, particularly DIVincome, are highly concentrated near 1, suggesting low variance. This lack of variation may weaken their statistical relevance in regression analysis.

In Fig. E.21, the boxplot reveals substantial differences in dispersion across the indices. DIVloan and DIVdeposits exhibit wider interquartile ranges, while DIVincome is tightly clustered. These differences may reflect varying degrees of heterogeneity across banks in terms of how they diversify specific balance sheet components.

Table E.20 reveals important distinctions in the distributional properties of diversification indices between defaulted and non-defaulted banks. While differences in means and medians are modest — e.g., slightly higher **DIVassets** and **DIVincome**, and lower **DIVloan** for defaulted banks — the higher-order moments differ substantially and significantly. Specifically, the standard deviation of **DIVdeposits** is markedly greater among defaulted banks (0.104 vs. 0.064), indicating more heterogeneity in deposit structures. Kurtosis and skewness metrics also show important contrasts: distributions for defaulted banks are generally less peaked and exhibit greater symmetry or left-skewness. For instance, **DIVassets** shows a transition from right-skewed (0.584) in non-defaulted banks to left-skewed (−0.497) in defaulted ones, and **DIVdeposits** kurtosis drops sharply from 14.40 to 7.97.

These findings suggest that while mean-level differences are not predictive on their own in logistic regressions, the shape and spread of the distributions differ significantly. This implies that nonlinear modeling approaches — potentially incorporating distributional asymmetries or tail behavior — may better capture the role of diversification in predicting bank default. Due to space constraints and the focus of the current study, this avenue is left for future research.

E.1.2. Long-term relationships: Johansen cointegration analysis

While the short-run predictive power of the indices appears weak, we turn to the Johansen cointegration test to assess whether the diversification indices are structurally tied to key bank-level financial variables—specifically, the interest-bearing assets-to-total assets ratio (I2.PN.TA) and return on assets (I4.ROA).

Table E.21 shows the trace statistics for various rank hypotheses. Each null hypothesis $r \leq k$ is strongly rejected at the 5% level, confirming the existence of multiple cointegrating vectors among the diversification indices and the financial indicators (see also Table E.22).

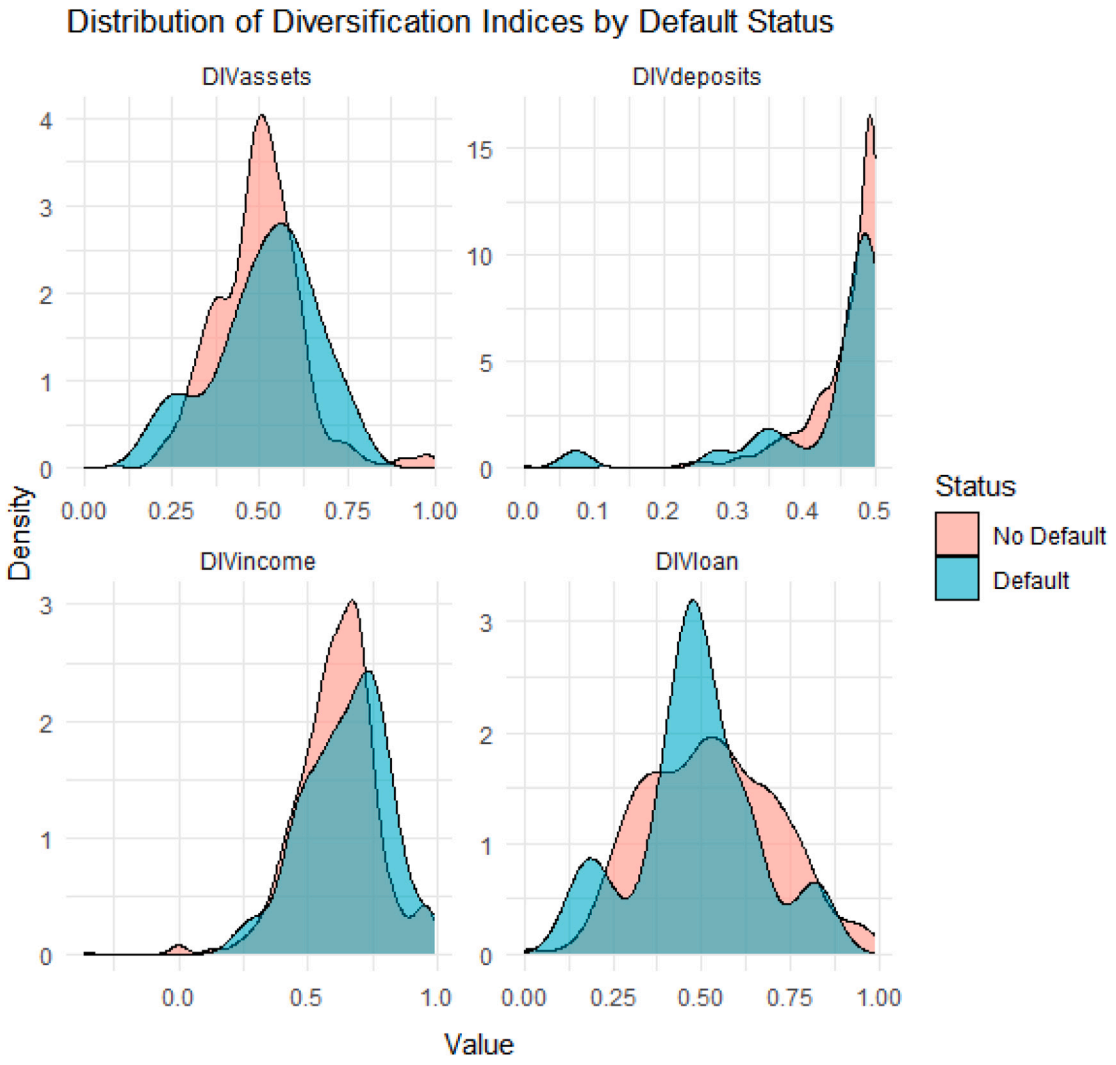


Fig. E.20. Distribution of Diversification Indices.

Table E.20
Distribution Moments of Diversification Indices by Default Status.

Statistic	DIVassets		DIVdeposits		DIVincome		DIVloan	
	Non-default	Default	Non-default	Default	Non-default	Default	Non-default	Default
Mean	0.497	0.517	0.453	0.435	0.612	0.651	0.528	0.486
Median	0.499	0.530	0.477	0.480	0.623	0.683	0.522	0.485
SD	0.128	0.148	0.064	0.104	0.156	0.164	0.185	0.174
Skewness	0.584	−0.497	−2.796	−2.286	−0.585	−0.226	0.124	−0.013
Kurtosis	5.245	2.634	14.401	7.966	5.618	2.866	2.490	3.044
Q25	0.418	0.438	0.435	0.433	0.524	0.571	0.382	0.429
Q75	0.566	0.603	0.495	0.494	0.703	0.757	0.668	0.584
Sample size (n)	990	21	990	21	990	21	990	21

[a] *Note:* Summary of the distributional moments of each diversification index by default status. Higher-order moments, such as skewness and kurtosis, highlight distributional differences that extend beyond mean-level comparisons.

To further interpret the long-term dynamics, Table E.23 displays the normalized cointegrating vectors. These coefficients represent stable long-run relationships between the diversification measures and the bank-specific indicators.

The adjustment dynamics of each variable in response to disequilibrium are reported in Table E.24. These loadings indicate how quickly deviations from equilibrium are corrected.

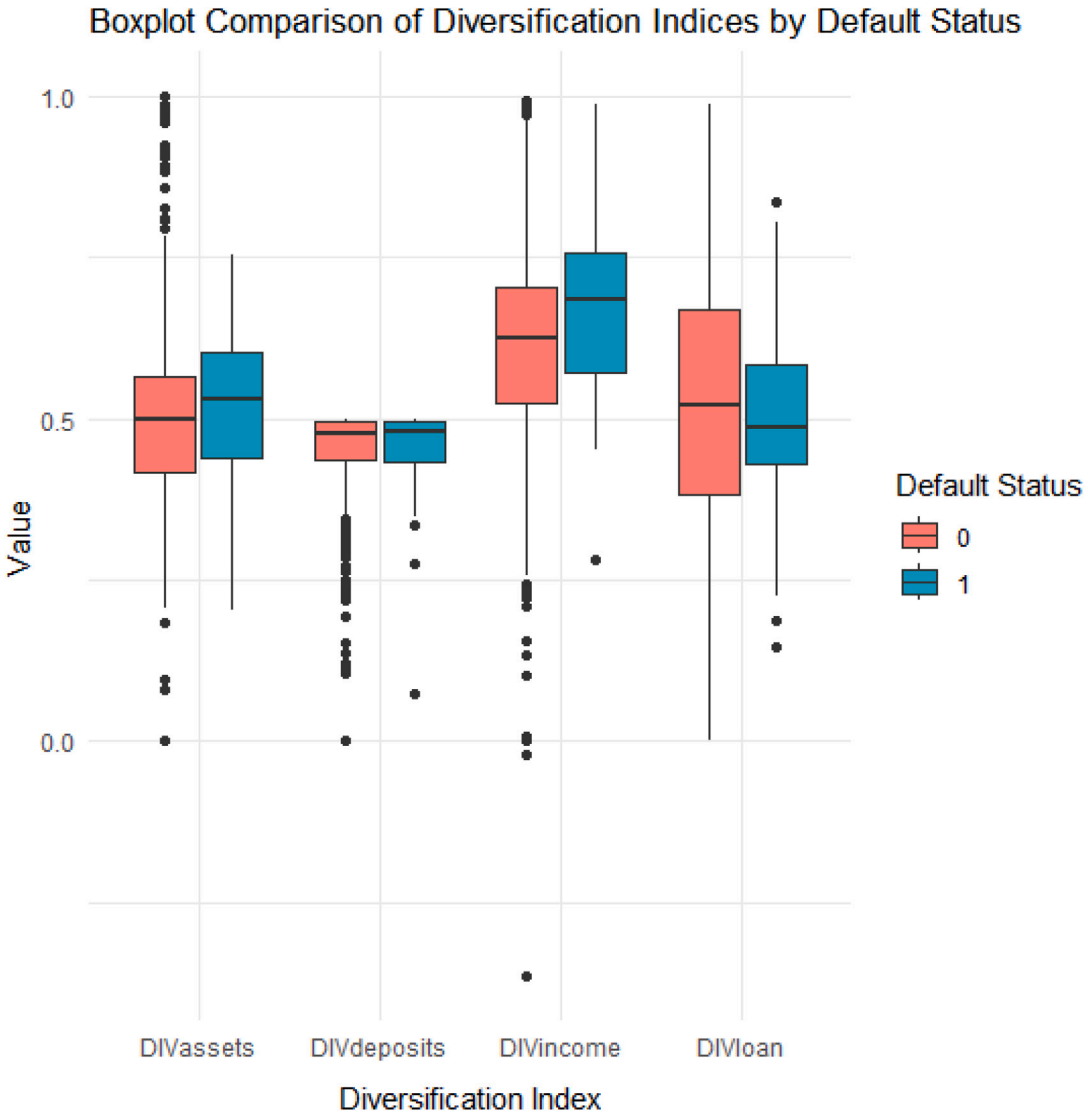


Fig. E.21. Boxplot Comparison of Diversification Indices.

Table E.21
Johansen Cointegration Test Results (Trace Statistic).

Hypothesis	Trace_Statistic	CV_10pct	CV_5pct	CV_1pct
r ≤ 1	851.47	71.86	76.07	84.45
r ≤ 2	611.12	49.65	53.12	60.16
r ≤ 3	391.64	32.00	34.91	41.07
r ≤ 4	223.98	17.85	19.96	24.60
r ≤ 5	85.80	7.52	9.24	12.97
r = 0	1151.44	97.18	102.14	111.01

Note: All trace statistics exceed their corresponding critical values, indicating that the null hypotheses of no or limited cointegrating relationships are rejected. This supports the presence of up to six long-run equilibrium relationships.

Table E.22
Johansen Cointegration Test Summary (Trace Test).

Null Hypothesis	Test Statistic	Critical Value (5%)	Decision	Interpretation	Comment
r ≤ 5	85.80	9.24	Reject H0	No cointegrating relations	Strong evidence
r ≤ 4	223.98	19.96	Reject H0	At most 1 relation	Strong evidence
r ≤ 3	391.64	34.91	Reject H0	At most 2 relations	Strong evidence
r ≤ 2	611.12	53.12	Reject H0	At most 3 relations	Strong evidence
r ≤ 1	851.47	76.07	Reject H0	At most 4 relations	Strong evidence
r = 0	1151.44	102.14	Reject H0	At most 5 relations	Strong evidence

Note: The summary table confirms the statistical rejection of all lower-order cointegrating ranks. The system likely features complex, multi-equation equilibrium dynamics.

Table E.23
Normalized Cointegrating Vectors.

	DIVassets.l2	DIVincome.l2	DIVdeposits.l2	DIVloan.l2	I2.PN.TA.l2	I4.ROA.l2	Constant
DIVassets.l2	1.000	1.000	1.000	1.000	1.000	1.000	1.000
DIVincome.l2	0.668	−0.954	1.444	−4.551	−0.071	0.005	−0.120
DIVdeposits.l2	1.577	1.467	3.887	2.796	−2.020	0.277	0.721
DIVloan.l2	−1.549	0.616	0.239	−0.589	−0.048	−0.805	−0.892
I2.PN.TA.l2	7.832	0.528	0.966	−2.755	−0.282	−0.896	−0.402
I4.ROA.l2	7.341	−30.039	29.425	74.363	7.805	−0.226	1.935
constant	−1.555	−0.903	−3.407	1.468	0.500	−0.120	4.977

Note: These cointegrating relationships suggest that long-run co-movements exist between diversification patterns and core bank performance metrics. The magnitude and direction of coefficients reflect differing equilibrium roles.

Table E.24
Loading Matrix: Speed of Adjustment.

	DIVassets.l2	DIVincome.l2	DIVdeposits.l2	DIVloan.l2	I2.PN.TA.l2	I4.ROA.l2	Constant
DIVassets.d	0.003	−0.084	−0.050	0.009	−0.145	−0.070	≈0
DIVincome.d	0.030	0.096	−0.051	0.045	−0.006	−0.039	≈0
DIVdeposits.d	0.009	−0.023	−0.042	−0.000	0.077	−0.009	≈0
DIVloan.d	0.074	−0.079	−0.075	0.013	−0.135	0.107	≈0
I2.PN.TA.d	−0.056	−0.013	0.008	−0.000	−0.025	0.043	≈0
I4.ROA.d	−0.002	0.006	−0.004	−0.002	−0.004	0.001	≈0

Note: Larger coefficients (e.g., for DIVloan.d and I2.PN.TA.d) imply quicker adjustments toward equilibrium. In contrast, I4.ROA.d changes slowly, indicating a more persistent deviation pattern.

E.1.3. Synthesis and implications

The findings highlight a clear distinction between the short-run predictive limitations and the long-run structural relevance of diversification. Logistic regression results show that diversification indices do not significantly improve default prediction and do not enhance the performance of the proposed parsimonious model. This suggests that, in the short term, diversification metrics offer limited incremental value beyond traditional financial ratios.

In contrast, cointegration analysis reveals that diversification indices are closely linked to key financial indicators over the long run, implying that they may still capture important equilibrium relationships. However, this analysis is based on the assumption of linear dynamics, as both the Johansen cointegration test and any resulting VECM rely on symmetric, linear adjustment mechanisms. This may mask nonlinearities, particularly during periods of financial stress or across heterogeneous banks. Due to space constraints and the scope of the present study, we leave the investigation of nonlinear alternatives — such as threshold cointegration or regime-switching models — for future research, which may better capture asymmetries and structural breaks in the role of diversification.

Appendix F. Macro variables analysis

F.1. Macro variables and Granger causality

To identify the most relevant macroeconomic predictors of bank default rates, Granger causality tests were applied to each indicator at lags 0, 1, and 2. Only variables demonstrating statistical significance at any lag were retained for further modeling, ensuring a parsimonious yet robust predictive framework. As shown in Table E.25, no significant effects were observed at lag 0 or lag 1. However, **CPI**, **GDP growth**, and **interest rates** exhibit significant Granger-causal relationships at **lag 2**. These findings suggest that macroeconomic shocks influence default risk with a delay, emphasizing the importance of incorporating lagged macroeconomic variables in predictive models to enhance forecast accuracy.

Table E.25
Granger Causality Test Results.

Variable (Lagged)	Lag	P_value	Significant ($p < 0.05$)
macro_CPI_IT	0.00	0.211	No
macro_LN.dGDP._IT	0.00	0.475	No
macro_IR_IT	0.00	0.551	No
macro_CPI_IT_lag1	1.00	0.227	No
macro_LN.dGDP._IT_lag1	1.00	0.061	No
macro_IR_IT_lag1	1.00	0.392	No
macro_CPI_IT_lag2	2.00	0.033	Yes
macro_LN.dGDP._IT_lag2	2.00	0.005	Yes
macro_IR_IT_lag2	2.00	0.000	Yes

Table F.26
Summary Parsimonious Logistic Model.

Coefficients	Estimate	Std. Error	z Value	Pr(> |z|)
(Intercept)	−3.7703	0.4374	−8.620	<2e-16***
macroCPIITlag2	−2.2351	3.9074	−0.572	0.5673
macroLNlogGDPITlag2	0.8396	0.3986	2.106	0.0352*
macroIRITlag2	1.4273	1.0345	1.380	0.1677

Signific. codes: *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$
Dispersion parameter for binomial family taken to be 1.
Null Deviance: 204.19 on 1008 degrees of freedom.
Residual Deviance: 197.82 on 1005 degrees of freedom.
AIC: 205.82
Observations removed due to missing values: 2
Note: This parsimonious logistic regression model shows that among the selected macroeconomic variables, only the lagged log-change in GDP (`macroLNlogGDPITlag2`) has a statistically significant effect on the outcome at the 5% level, while the overall model exhibits a moderate fit (AIC = 205.82).

The positive coefficient on the second lag of GDP growth suggests that an economic expansion may precede, with a two-period delay, an increase in the probability of default. This aligns with the 'boom and bust' effect: a period of economic expansion leads to credit expansion, which finances overproduction. This, in turn, exposes companies to post-growth fragility, with a subsequent reduction in production and deterioration of assets.

F.2. Logistic regression

Table F.26 presents the results of the estimated logistic regression using only those macroeconomic variables that were found to be significant in the Granger causality analysis. Among the included predictors, only the **lagged log-change in GDP** (`macroLNlogGDPITlag2`) is statistically significant at the 5% level ($p = 0.0352$), indicating that increases in GDP growth reduce the likelihood of bank default. Although the coefficients for lagged CPI and interest rates have the expected signs, their effects are not statistically significant. The model exhibits a reasonable fit, with a residual deviance of 197.82 and an AIC of 205.82, suggesting improved explanatory power relative to the null model. These findings underscore the relevance of lagged macroeconomic conditions — particularly GDP growth — as predictors of default risk.

The residuals analysis shown in Fig. F.22 highlights potential issues with the model fit, including the presence of outliers and indications of non-linear relationships between predictors and the outcome variable.

F.2.1. Interaction and nonlinear effects analysis

To further investigate the influence of firm-level and macroeconomic variables on bank default probability, we extended the baseline logistic regression by incorporating interaction and nonlinear effects. Specifically, a logistic regression model with an interaction term between firm profitability (`I4.ROA`) and lagged GDP growth (`macro_LN.dGDP._IT_Lag2`) was estimated (Table F.27). The results reveal that `I4.ROA` exerts a strong and statistically significant negative effect on default likelihood, consistent with theoretical expectations that more profitable banks are less likely to default. In contrast, the lagged GDP growth variable and its interaction with `I4.ROA` were not statistically significant, suggesting that the impact of macroeconomic growth does not vary systematically with profitability in a linear framework.

Given the evidence of potential model misspecification from residual diagnostics, we further estimated a Generalized Additive Model (GAM) allowing for a smooth, nonlinear interaction between `I4.ROA` and `macro_LN.dGDP._IT_Lag2` (Table F.28). Despite the model explaining a substantial proportion of deviance (71.5%) and showing an adjusted R^2 of 0.605, the smooth interaction term was not statistically significant ($p = 0.757$). These findings suggest that neither linear nor nonlinear interactions between firm profitability and lagged GDP growth meaningfully enhance the prediction of default risk beyond the main effects already captured in prior models.

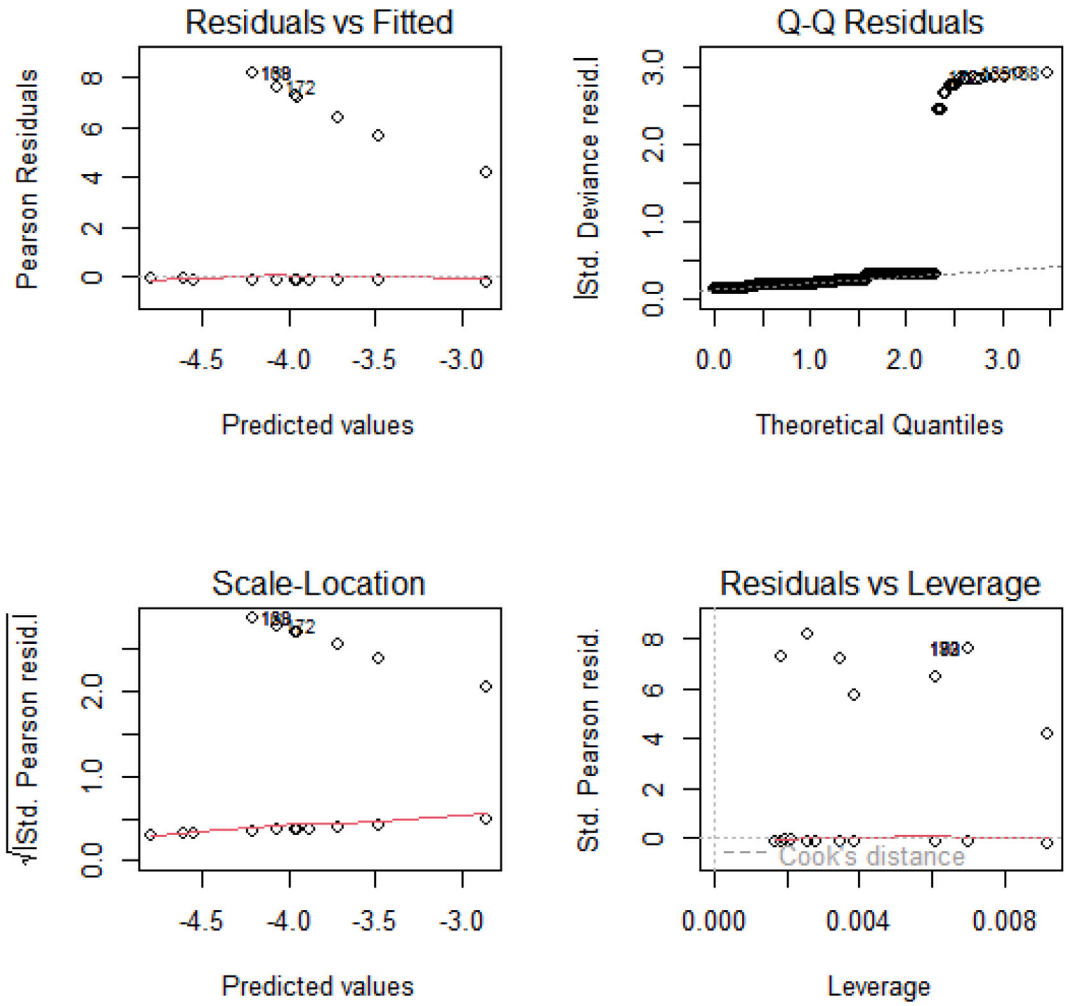


Fig. F.22. The analysis of residuals indicates that the model may suffer from the presence of outliers and potential non-linearity in the relationship between predictors and the outcome.

Table F.27
Summary Logistic Model With Interaction.

Coefficients	Estimate	Std. Error	z Value	Pr(> |z|)
(Intercept)	−3.688	0.839	−4.396	1.10e−05***
I4.ROA	−159.318	37.199	−4.283	1.85e−05***
macro_LN.dGDP._IT_Lag2	0.156	0.914	0.171	0.864
I2.PN.TA	−12.911	9.114	−1.417	0.157
BCC	NA	NA	NA	NA
I4.ROA:macro_LN.dGDP._IT_Lag2	−11.085	65.651	−0.169	0.866

Signific. codes: $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$
Dispersione parameter for binomial family taken to be 1.
Null Deviance: 77.316 on 647 degrees of freedom.
Residual Deviance: 59.165 on 643 degrees of freedom.
AIC: 69.165
Number of Fisher Scoring Interactions: 8.

F.3. Stress testing framework: Scenario design and interaction effects

We define three hypothetical stress scenarios — mild, moderate, and severe — based on empirical quantiles of key explanatory variables, including bank profitability, capitalization, and macroeconomic growth. Table F.29 outlines the quantile thresholds and their corresponding economic interpretations.

Table F.28
Summary of GAM Logistic Model with Smooth Interaction Term.

Coefficients	Estimate	Std. Error	z Value	Pr(> |z|)
(Intercept)	0.000	0.000	Nan	Nan
I2.PN.TA	−0.035	8.398	−0.004	0.997
BCC	−199.238	1616.346	−1.123	0.902
Smooth Term (approximate significance):				
`s(I4.ROA, macro_LN.dGDP._IT_Lag2)`: edf = 10.44, Ref.df = 11.27, χ^2 = 6.793, p-value = 0.757.				

Signific. codes: $^{*}p < 0.05$; $^{**}p < 0.01$; $^{***}p < 0.001$.
R-squared (adj.) = 0.605; Deviance explained = 71.5%.
UBRE = −0.92761; Scale estimate = 1; n = 648.

Table F.29
Definition of Stress Scenarios and Corresponding Quantiles.

	Scenario	Quantile	Description
1	Mild Adverse Scenario	0.25	Minor market or macroeconomic disruptions
2	Moderate Stress Scenario	0.15	Significant financial and economic strain
3	Severe Shock Scenario	0.05	Severe systemic crisis or extreme shock

Table F.30
Descriptive Moments of ROA × GDP Lag 2, by Default Status.

Status	Mean	Median	Variance	Skewness	Kurtosis	IQR_{25}	IQR_{75}
0	0.00009	−0.00000	0.00001	−1.766	20.998	−0.00075	0.00119
1	−0.00307	0.00038	0.00012	−0.116	−1.960	−0.01218	0.00517

Note: Statistics are segmented by default status. Status = 1 denotes defaulting banks.

F.3.1. Interaction effects of profitability and growth under stress

To capture interaction effects between bank-specific performance and macroeconomic conditions, we construct the variable ROA ×GDP Lag 2 as the element-wise product of return on assets and lagged GDP growth. This metric reflects how profitability interacts with broader economic activity during stress events.

Descriptive statistics for this interaction term, stratified by default status, are shown in Table F.30. Defaulting banks (status = 1) exhibit lower means and higher variance, along with a more extreme interquartile range (IQR), suggesting greater sensitivity to joint profitability-growth shocks.

Fig. F.23 further illustrates these differences by displaying the full distribution of the interaction term. The distribution for defaulting banks is notably wider and shifted, supporting the descriptive findings. The corresponding boxplot in Fig. F.24 highlights this with clearer IQR spread and outlier behavior, confirming the greater dispersion and potential downside risk associated with defaulting institutions.

F.4. Scenario-specific stress quantiles by bank group

This section presents stress scenario quantiles for key risk drivers, computed separately for Cooperative Credit Banks (BCC = 1) and non-cooperative banks (BCC = 0). Tables F.31 (BCC = 0) and F.32 (BCC = 1) summarize these quantiles across mild, moderate, and severe stress scenarios for four variables: ROA, lagged GDP growth (GDP Lag 2), equity-to-assets ratio, and the interaction term ROA ×GDP Lag 2.

The quantile profiles suggest marked heterogeneity in stress responses by bank type. Cooperative Credit Banks (BCC = 1) consistently show higher ROA values under all scenarios, indicating stronger profitability buffers. For instance, under the severe shock scenario, BCC 1 banks exhibit a less negative ROA (−0.005) compared to BCC 0 banks (−0.017). Similarly, the equity-to-assets ratio is generally higher for BCC 1 under mild and moderate stress, although this difference narrows in the severe case. These differences suggest that BCC 1 banks may be more resilient under baseline and moderate conditions but are not immune to tail risks.

GDP Lag 2, a systemic macroeconomic driver, remains constant across bank types — as expected — since it reflects external conditions. However, the interaction term ROA ×GDP Lag 2 shows more favorable values for BCC 1 in every scenario. This implies that the joint impact of earnings and economic deterioration is less severe for cooperative banks, highlighting structural differences in sensitivity to macro-financial feedback loops.

Figs. F.25 and F.26 provide visual confirmation of these patterns. Non-cooperative banks (BCC = 0) display steeper declines in ROA and equity ratios as stress intensifies, whereas the plots for BCC 1 show more stable trajectories. The interaction term also appears more volatile for BCC 0, indicating a higher degree of exposure to macro-financial amplification.

Together, these results reinforce the importance of accounting for bank-type heterogeneity in the design of stress testing frameworks and capital adequacy assessments.

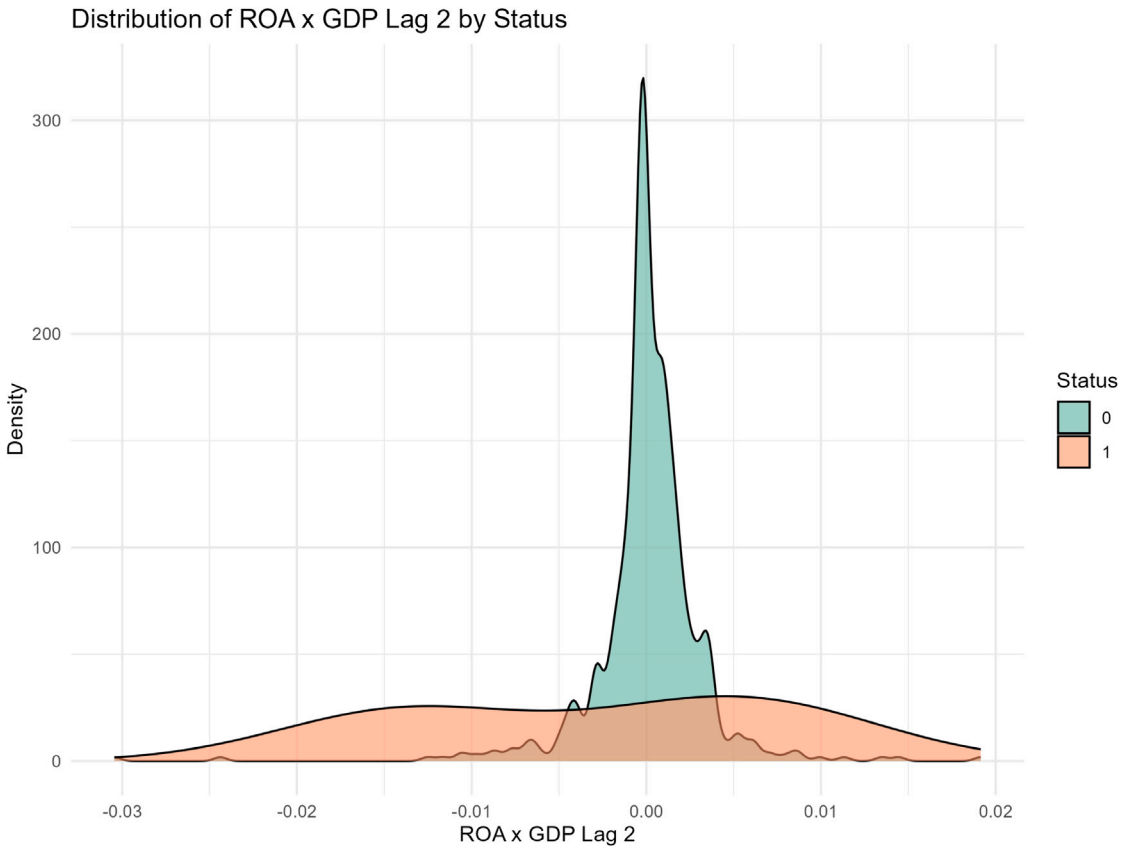


Fig. F.23. Distribution of ROA × GDP Lag 2 by Default Status.

Table F.31
Stress Quantiles for BCC = 0 (Non-Cooperative Banks).

	Variable	Mild Adverse	Moderate Stress	Severe Shock	Shock Sign	Interpretation
1	ROA	0.00000	−0.00500	−0.01700	–	Lower Profitability
2	GDP Lag 2	−0.37300	−0.70900	−0.96300	–	Weaker Economic Growth
3	Equity/Total Assets	0.06500	0.05900	0.04600	–	Higher Financial Leverage
4	ROA × GDP Lag 2	−0.00112	−0.00213	−0.00481	–	Joint Impact of Low Profitability and Weak Growth

Table F.32
Stress Quantiles for BCC = 1 (Cooperative Credit Banks).

	Variable	Mild Adverse	Moderate Stress	Severe Shock	Shock Sign	Interpretation
1	ROA	0.00100	0.00100	−0.00500	–	Lower Profitability
2	GDP Lag 2	−0.37300	−0.70900	−0.96300	–	Weaker Economic Growth
3	Equity/Total Assets	0.07200	0.06400	0.04500	–	Higher Financial Leverage
4	ROA × GDP Lag 2	−0.00075	−0.00186	−0.00432	–	Joint Effect of Low Profitability and Weak Growth

F.5. Comparative analysis of stress responses across bank types

Table F.33 summarizes the intergroup differences in stress quantiles between Cooperative Credit Banks (BCC = 1) and other banks (BCC = 0). Capital ratios are moderately stronger for BCC 1 under mild and moderate stress, but slightly weaker under severe conditions. ROA remains higher for BCC 1 across all scenarios, suggesting relative resilience. The interaction term also consistently favors BCC 1, reinforcing the structural variation in stress tolerance.

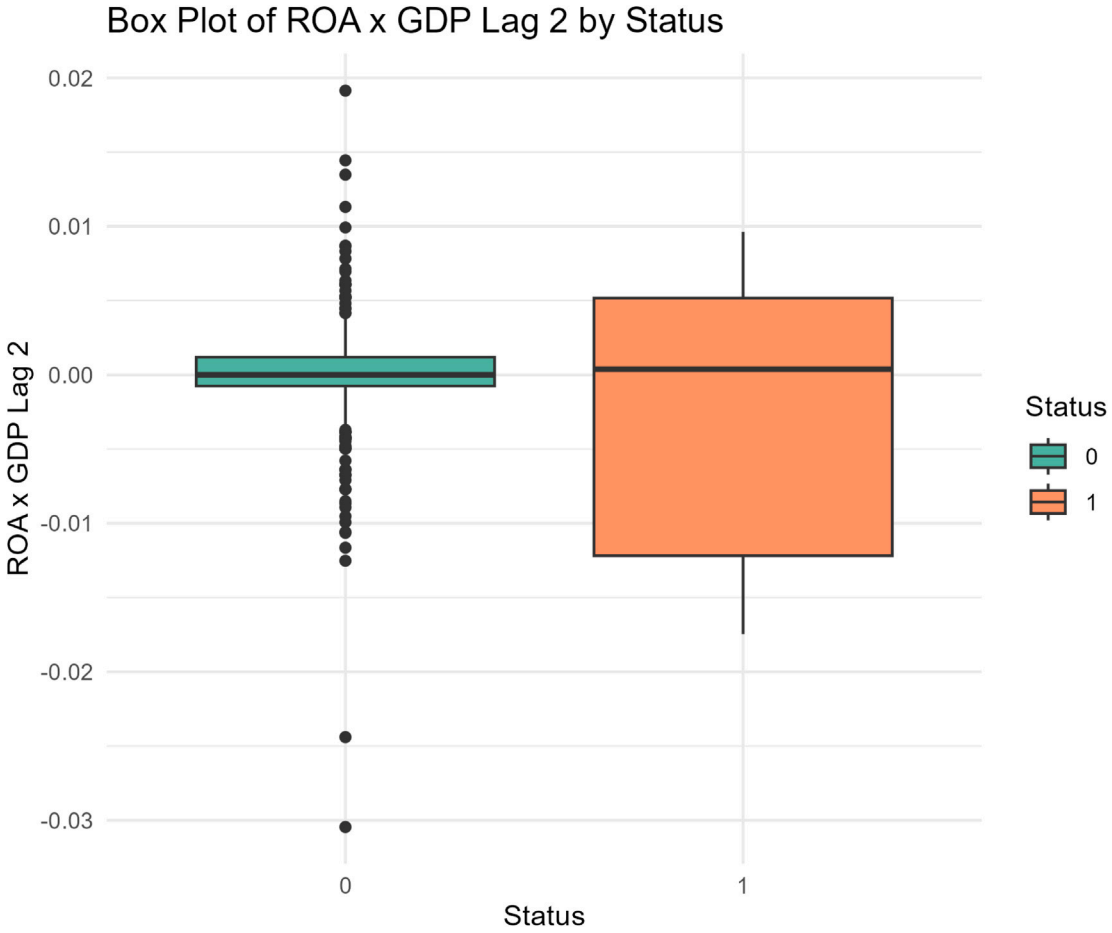
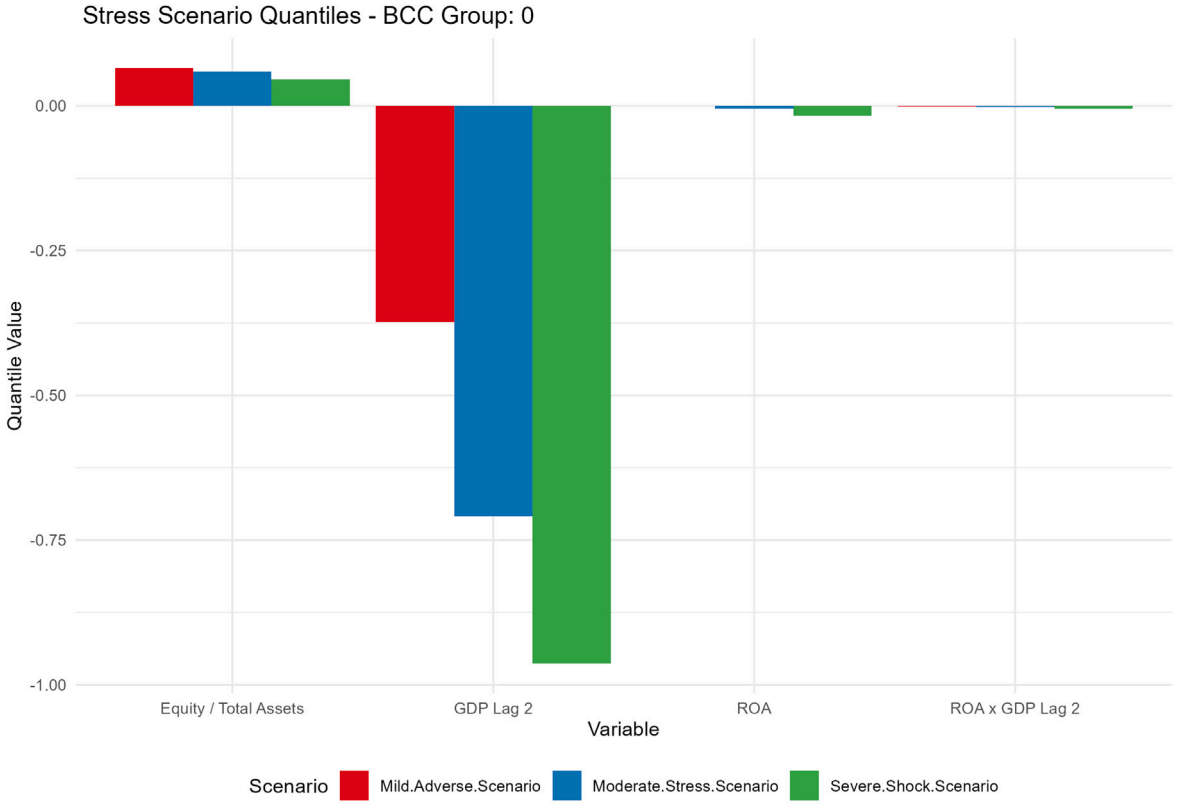


Fig. F.24. Boxplot of ROA ×GDP Lag 2 by Default Status.

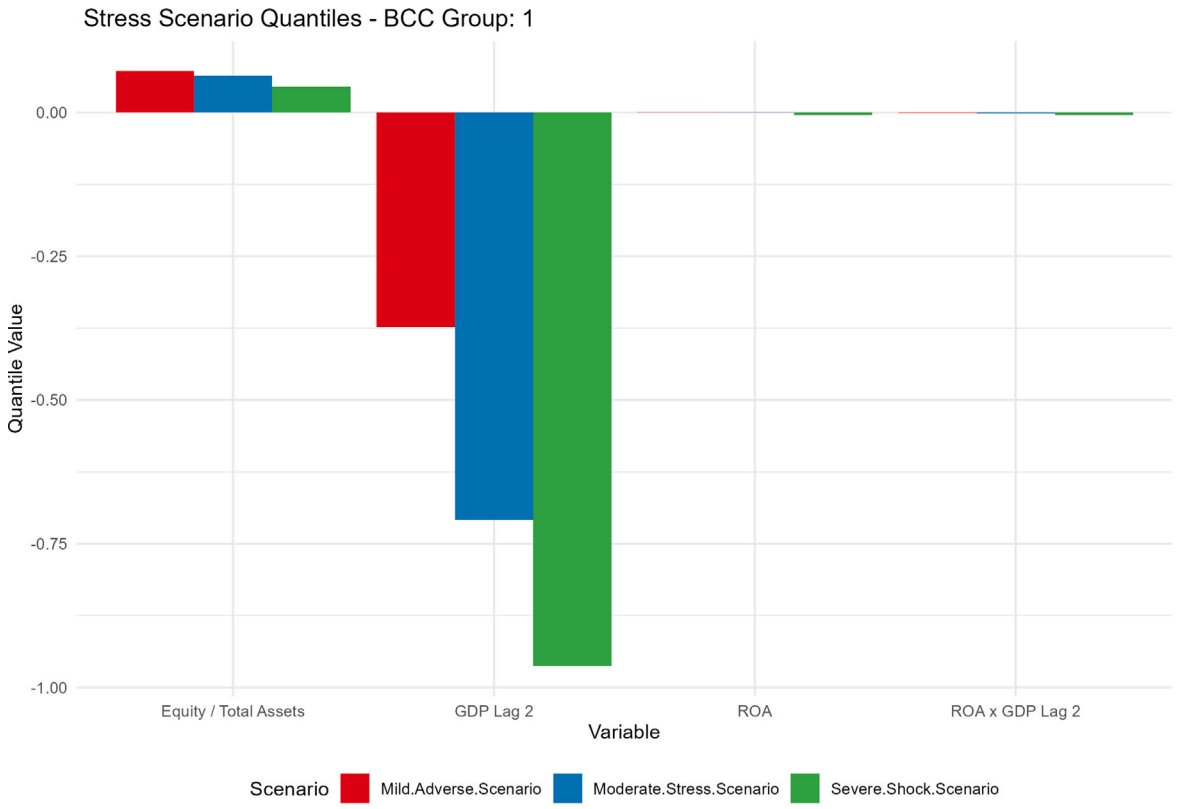
Table F.33
Intergroup Differences in Stress Quantiles (BCC 1 minus BCC 0).

	Variable	Mild Adverse	Moderate Stress	Severe Shock
1	Equity/Total Assets	0.00700	0.00500	−0.00100
2	GDP Lag 2	0.00000	0.00000	0.00000
3	ROA	0.00100	0.00600	0.01200
4	ROA × GDP Lag 2	0.00037	0.00027	0.00050



Note: Quantiles represent the stress values for each scenario.
- ROA (Return on Assets): lower quantiles indicate lower profitability (negative stress).
- GDP Lag 2 (2-period lagged GDP growth): lower quantiles indicate weaker economic growth (negative stress).
- Equity / Total Assets: lower quantiles indicate lower capital strength/higher leverage (negative stress).
- ROA x GDP Lag 2: lower quantiles indicate a combination of lower profitability and weaker economic growth (negative stress).

Fig. F.25. Stress Scenario Impacts for BCC = 0 (Non-Cooperative Banks).



Note: Quantiles represent the stress values for each scenario.
- ROA (Return on Assets): lower quantiles indicate lower profitability (negative stress).
- GDP Lag 2 (2-period lagged GDP growth): lower quantiles indicate weaker economic growth (negative stress).
- Equity / Total Assets: lower quantiles indicate lower capital strength/higher leverage (negative stress).
- ROA x GDP Lag 2: lower quantiles indicate a combination of lower profitability and weaker economic growth (negative stress).

Fig. F.26. Stress Scenario Impacts for BCC = 1 (Cooperative Credit Banks).

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